Comparison of the precision of different ML architectures for defect detection in guided wave signals under varying EOCs

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Summary

The suitability of two deep learning models is assessed in the context of guided wave structural health monitoring under varying environmental and operational conditions. The traditional optimal baseline subtraction model is implemented as a comparison benchmark, which produces a pristine trace through baseline manipulation. A long short-term memory network with recurrent memory and a convolutional neural network autoencoder are trained using healthy guided waves to construct an equivalent pristine response. Subtraction of the two traces produces a residual used for defect detection. The three models are evaluated using 19.5 months of experimental data collected from an array of guided wave sensors on a steel water tank. Synthetic defects are superimposed onto healthy traces for controlled defect amplitudes and arrival times allowing detection performance to be compared using receiver operating characteristic curves. Results from synthetically generated defects show consistent and substantial improvement of the LSTM over the traditional approach. Additional experimental data were collected with the aim of future model testing and defect localisation, although reliable localisation is not achieved in this study. Overall, the LSTM showed promising results from synthetic testing, but further testing using multiple sensor paths and real defects should be performed.

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Declaration

This project report is submitted towards an application for a degree at the University of Bristol. The report is based upon independent work by the candidates. All contributions from others have been acknowledged and the supervisor is identified on the front page. The views expressed within the report are those of the authors and not of the University of Bristol.

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Work Allocation

	ES	OD	RJ	BT	LH
Introduction	●	●			
Fundamentals of Guided Wave Damage Identification	●	●	●	●	
General overview				●	
Baseline subtraction		●			
Influence of temperature on guided wave signals		●			
OBS		●			
CNN			●		
LSTM architecture description	●				
Evaluation metrics	●				
Data Collection		●			
Numerical Methods	●	●	●		
OBS		●			
CNN			●		
LSTM	●				
Synthetic Defect Detection Results	●	●	●		
Experimental Testing				●	●
Experimental Results		●		●	●
Discussion	●	●	●	●	
Conclusion	●	●			
Report formatting and editing	●	●	●		

Key

● : Group member has contributed the majority/all of the work for this section.

● : Group member has shared contribution of the work for this section.

● : Group member has provided assistance for this section.

We confirm that the information on this page accurately describes our individual contributions to the project.



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1 Introduction (OD, ES)

The detection of damage inside a structure is an extensively researched challenge across the full spectrum of engineering disciplines. There has been an increasing paradigm shift away from traditional non-destructive evaluation (NDE) techniques, which involve taking the structure of interest out of operation to attach sensors temporarily, and towards structural health monitoring (SHM). SHM utilises permanently attached sensors capable of continuous monitoring of the structure and is able to overcome many of the limitations of NDE, namely its restriction to localised inspections, the need for structural disassembly and its large dependence on human interpretation [1]. This is significant as it can heavily reduce maintenance costs by minimising downtime and enabling defects to be detected and monitored from their first appearance, thus increasing the design life of the structure [2]. Within the field of SHM, guided wave structural health monitoring (GWSHM) has attracted significant attention since ultrasonic guided waves can provide large-area inspection coverage of plate-like structures using only a sparse array of sensors [3].

This project investigates residual-based guided wave defect identification under realistic environmental and operational conditions (EOCs). Section 1.1 details the relevant engineering background and the limitations of existing approaches, then Section 1.2 goes on to define the project objectives and the contributions made in this study.

1.1 Background (OD, ES)

The application of GWSHM within a range of different industries has been recently investigated, including damage detection in pipeline structures [4], hydrogen transportation systems [5] and aerospace applications [6]. There has also been a particular focus on upscaling the technique for use on industrial-size composite structures [7, 8]. The GWSHM systems used in this project comprise transducer pairs operating in pitch-catch mode, in which one transducer actuates the guided wave and the other records the structural response. Lamb waves, a type of ultrasonic guided wave that propagate through thin, plate-like structures, are used due to their long propagation distances, low attenuation speeds and high sensitivity to defects [9]. These characteristics have been shown to be conducive towards lower power consumption and thus are a cost-efficient choice for SHM [10, 11].

Defect detection is inherently difficult due to the complexity of the measured signals; a typical time-domain response (time-trace) contains overlapping wavepackets from direct transmission and from reflections from benign structural features such as edges and flanges [3]. Since damage reflections are relatively weak (primarily observed between -15 and -40 dB relative to edge reflections) [12,13] they can be easily masked by noise in the signal. As a result, defect detection relies fundamentally on baseline subtraction in which a pristine signal from the structure is subtracted from the current measurement, leaving what is termed the residual signal; since the structural features remain unchanged, they are reduced significantly (or removed entirely in the ideal case) on subtraction [14].

The key limitation of baseline subtraction as an approach lies in the difficulty in differentiating between residuals caused by damage in the structure and residuals caused by changing EOCs. In particular, changes in temperature have been shown to cause significant, unavoidable repeatability issues in the recorded signals. Firstly, temperature variations cause thermal expansion/contraction of the structure which physically changes the propagation distance between fixed sensors, alongside changing the material's density and thickness [13, 14]. Secondly, the elastic moduli of the structure are altered, which shifts the phase velocity of the signal and leaves a residual largely governed by the time-shift of waves within the wavepacket [2, 14]. Temperature can also influence the frequency performance of transducers, however this effect is generally smaller than propagation-related changes for modest variations in temperature [14].

A widely used benchmark approach for resolving these distortions is optimal baseline subtraction (OBS), and was first proposed by Lu and Michaels [15]. This method uses a library of signals recorded under

varying EOCs from a time when the structure was known to be defect-free. The current measurement is matched to the closest stored baseline signal and then subject to optimised amplitude scaling and time dilation to improve the match [2]. OBS is attractive due to being physically interpretable and only requiring healthy data. It is limited, however, by its requirement for a baseline library spanning the full range of EOCs the structure is expected to endure in its lifetime in order to achieve sufficient temperature resolution. This is challenging in long-term monitoring scenarios where seasonal cycles, sensor ageing and structural degradation could move the structure outside of the original baseline range. In addition, OBS applies global transformations to the entire baseline signal, implicitly assuming that EOCs cause consistent global distortions of the guided wave signals. In practice, different wavepackets can be affected differently by local temperature gradients and scattering paths. Consequently, the use of a single amplitude scale and time delay may align one part of the signal whilst other regions are left mismatched. This causes the accuracy of the method to be compromised under non-homogeneous temperature changes, since to account for these local distortions the baseline library would need to span an unattainable multidimensional parameter space [3]. To provide a comparison benchmark, the OBS method is applied to the data in this project.

Existing work on SHM has moved beyond OBS and begun investigation into a range of data-driven machine learning (ML) methods for defect detection and monitoring. Cha et al. [16] provide a broad review of multiple methods of deep learning implementation in SHM, including vibration-based monitoring, guided-wave inspection, data recovery and physics-informed modelling. More specifically within GWSHM, Yang et al. [17] discuss guided-wave propagation modelling, data acquisition, preprocessing, ML modelling and physics-based ML. Ying et al. [18] used adaptive boosting and a support vector machine for feature-based damage classification under varied EOCs in a steel pipe. Similarly, Liu and Zhang [19] designed an image-classification convolutional neural network (CNN) to detect damage in thin plates, done by applying a short time Fourier transform (STFT) of the signal data to create spectrum images of the guided waves. Most recently, Wu et al. [20] moved toward a time-frequency image representation, rather than a manually described feature one, performed through the use of a continuous wavelet transform alongside a deep CNN for Lamb wave damage detection in composite structures. These studies demonstrate the potential of machine learning in an SHM setting under limited EOC variation, supporting the feasibility of deep learning for this purpose; however, they also highlight several practical limitations [3]. Most significantly, supervised classifier-based approaches require an extensive range of labelled defect sizes, orientations, types and locations across a broad range of EOCs to provide sufficient coverage of the failure modes and operating condition space. Acquiring these defective datasets is both costly and labour intensive, potentially requiring the intentional damage of expensive equipment or structures, making purely supervised methods less feasible in an industrial setting.

This limitation motivated the 2022 study by Wang et al. [3], which investigated a trace-reconstruction predictive model for guided wave defect detection. A nonlinear autoregressive exogenous (NARX) architecture was trained on a set of healthy signal data in order to estimate an expected healthy response. This prediction could be subtracted from the real signal to form a residual in a similar manner to OBS methods. This residual could then be assessed using a damage index and threshold to determine whether a structure is likely to contain a defect. Predictive modelling like the NARX can be trained using only healthy data across the range of expected EOCs, which is much easier and far less costly to obtain, making it a more scalable and practically attractive method for defect detection.

Wang et al. [3] showed that a single-step NARX predictor could improve upon OBS by reconstructing the expected pristine signal, rather than relying on a close baseline match. The success of this model over traditional methods demonstrated the feasibility of using healthy-data prediction for residual-based guided wave defect detection. However, Tu et al. [21] later identified limitations in the single step NARX predictor and found inconsistent detection across different defect arrival times. This was theorised to be a result of the simplicity of the single-step prediction task. Specifically, this NARX model was trained

to minimise next-step prediction error, but low MSE did not necessarily correspond to consistent defect detection. It was suggested that, instead of truly learning the broader temporal structure governing the waveform evolution, the model could rely on local waveform extrapolation. This allowed defects to be partially reconstructed in the prediction rather than remaining clearly visible in the residual, reducing both the sensitivity and consistency of defect detection.

Although the most recent studies have explored a range of deep learning approaches to SHM, it is still noted by Cha et al. [16] that, "DL-based SHM appears to be in its infant stages" and many still require large labelled defective datasets or have only been validated under limited EOC variation. Consequently, there is a research gap that is addressed in this project: there remains a need for residual construction methods that retain the practical advantage of OBS by training only with healthy data, whilst being able to form defect-sensitive residuals that remain robust under EOC variation. This motivates the comparison of the traditional OBS method with two healthy data driven ML approaches that are novel to this purpose: a CNN autoencoder and an LSTM predictor.

1.2 Project objectives (OD, ES)

The overall aim of this project is to evaluate whether healthy data-driven reconstruction and prediction methods can improve guided wave defect identification under realistic EOC variation in comparison to the traditional OBS.

The hypothesis of this study is that both data-driven trace prediction models will outperform classical OBS trace fitting for residual-based defect detection. On top of this hypothesis, it is expected that the learned recurrent memory of the LSTM will allow temporal dependencies to be captured effectively, potentially providing improved prediction accuracy and defect sensitivity over the explicit fixed lag structure of the single-step NARX predictor [3].

To investigate this hypothesis, the objectives and contributions of the project are as follows:

1. To analyse a long-term dataset from an existing water tank monitoring setup and characterise the effect of naturally occurring EOCs on the measured guided wave signals
2. To generate a controlled synthetic defect dataset using experimentally measured pristine traces from the tank for detection performance to be assessed across known defect amplitudes and arrival times
3. To implement OBS as a comparison benchmark for residual-based guided wave defect detection
4. To develop and evaluate two different data-driven ML approaches for guided wave subtraction: a CNN autoencoder and an LSTM predictor
5. To compare the detection performance of OBS, CNN and LSTM using receiver operating characteristic (ROC) curves and area under the curve (AUC) metrics
6. To conduct a laboratory plate experiment to simulate a defect in the structure using magnets and analyse the resulting signals using a probabilistic data fusion method and the delay-and-sum (DAS) technique to localise the defect

Comparisons of the CNN and LSTM to the NARX model proposed by Wang et al. [3] are also discussed qualitatively, as a direct numerical comparison cannot be claimed due to the models being trained and tested on different sensor pairs from the same sensor network.

2 Fundamentals of Guided Wave Damage Identification (BT, OD, RJ, ES)

2.1 General overview (BT)

Guided waves are ultrasonic waves that propagate throughout a structure, guided by its boundaries. Lamb waves are the predominant type of guided ultrasonic stress wave used to evaluate plates or shells, although viable alternatives such as shear horizontal waves can be practical in specific cases despite actuation difficulties [22]. Lamb waves are highly desirable in SHM due to their high propagation distance, ability to detect internal damage via full cross-section coverage and high sensitivity to defects, but are dispersive, meaning wave velocity varies with frequency. This leads to the distortion of wave packets and reduction in wave amplitude over long distances.

Lamb wave propagation can be separated into its asymmetric (A_0) and symmetric (S_0) fundamental wave modes, where flexural motion occurs in the asymmetric mode and the symmetric mode is dominated by longitudinal motion, as demonstrated in Figure 1.

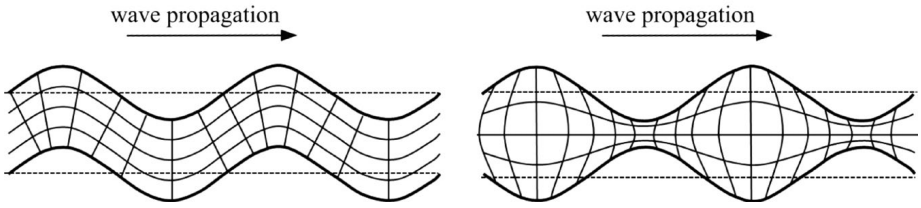
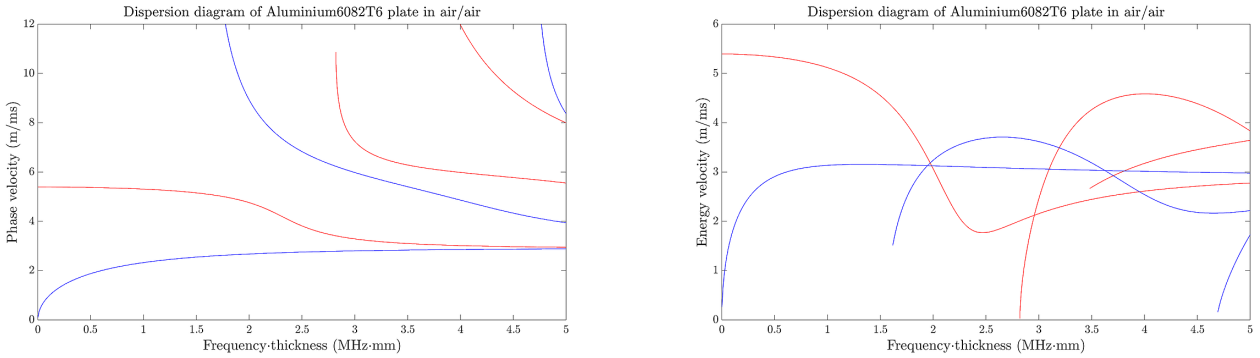


Figure 1: Representation of fundamental Lamb waves in thin plate-like structures with boundaries represented by dashed lines: antisymmetric A_0 mode (left) and symmetric S_0 mode (right), from Marks et al. [23]

Higher order modes appear at a greater frequency-thickness product; this is usually not desirable due to the added complexity it introduces to the signal, but have seen some applications, notably the zero-group-velocity Lamb waves most easily excited at S_1. Nevertheless, the fundamental modes are usually the modes of evaluation. Given the in-plane propagation of the S_0 mode, it is much more suited than the A_0 mode in applications that involve contact with liquid, such as the evaluation of water tanks, although it is noted that liquid was not present in the original water tank setup. The mathematical representation of Lamb waves is given by the transcendental Rayleigh-Lamb equation, which establishes the relationship between the excitation angular frequency and phase velocity. The equation can be used to plot the Lamb wave phase velocity dispersion curve as seen in Figure 2, which shows the variation of velocity of a single frequency component against the frequency-thickness product. The group velocity (sometimes also named the energy velocity), which represents the estimated velocity of the entire wave packet can also be plotted using the Rayleigh-Lamb equation.

To transmit the aforementioned guided waves, piezoelectric transducers are the widely accepted choice for GWSHM wave actuation and sensing due to their electromechanical coupling capability enabling bidirectional conversion between mechanical stress and electrical charge. While lead zirconate titanate (PZT) is traditionally considered as the leading piezo technology, polyvinylidene fluoride (PVDF) has emerged in recent years as potential option. While earlier literature [24] states that PZT is the more popular material among GWSHM researchers due to the reduced capabilities of PVDF, newer literature [25] successfully demonstrated the growing viability of PVDF transducers.

In practice, piezos typically take the form of inexpensive surface-bonded or embedded wafers configured into a network following one of two main strategies: the pitch-catch setup, where distinct piezo actuator-sensor pairs evaluate wave transmission, and the pulse-echo setup, where a single piezo is used to both transmit waves and receive subsequent reflected signals. These are shown visually in Figure 3.



(a) **(b)**

Figure 2: Lamb wave dispersion curves through 6082T6 Aluminium plotted using Dispersion Calculator by Armin Huber DLR, showing (a)Phase velocity plotted against frequency-thickness product, and (b)Group velocity plotted against frequency-thickness product

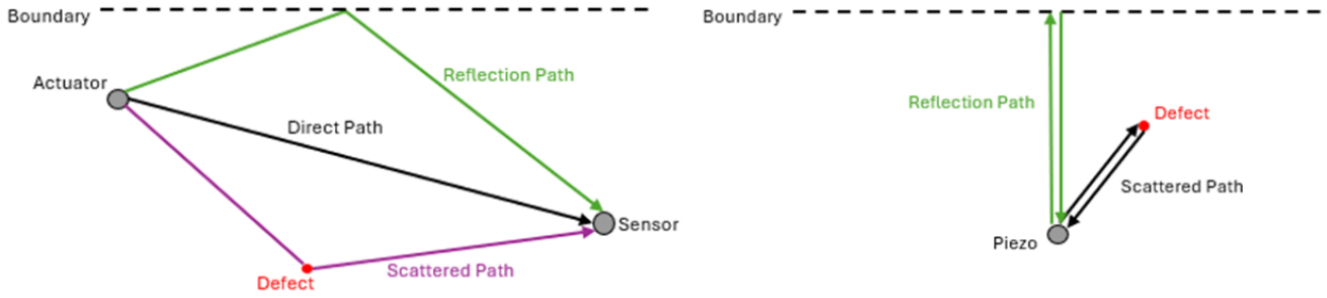


Figure 3: Diagrams showing the configuration of pitch-catch mode (left) and pulse-echo mode (right).

2.2 Residual-based defect detection (OD)

The foundational concept to the three models explored in this project is the generation of a resultant time-trace through comparing an experimentally measured time-trace from the structure, $s(t)$, against an estimate of the corresponding pristine signal, $\hat{s}(t)$, shown by

$$e(t) = s(t) - \hat{s}(t) \tag{1}$$

where $e(t)$ is what is termed the residual signal time-trace, containing information on any defects that appear in the structure during its operational life. In the ideal case, $e(t)$ is equal to zero if no reflections associated with damage are present in the signal and $\hat{s}(t)$ is perfectly estimated. In OBS, $\hat{s}(t)$ is produced through transformation of the best matched signal in the baseline library. For the CNN and LSTM, $\hat{s}(t)$ is reconstructed or predicted by a model trained with healthy data. A schematic diagram of the overall process of residual-based defect detection is shown in Figure 4.

2.3 Influence of temperature on guided wave signals (OD)

Temperature variation has been identified as the dominant environmental factor affecting the long-term stability of GWSHM [2]. For residual-based approaches, damage is interpreted from $e(t)$ resulting from the subtraction in Equation 1, which is fundamentally reliant on the assumption that any residual components above a determined threshold are being caused by reflections associated with damage to the structure. In practice, however, changes in temperature alter the waveform of guided wave signals. This produces residual components that can mask or mimic defects, particularly those of weaker amplitudes. The three

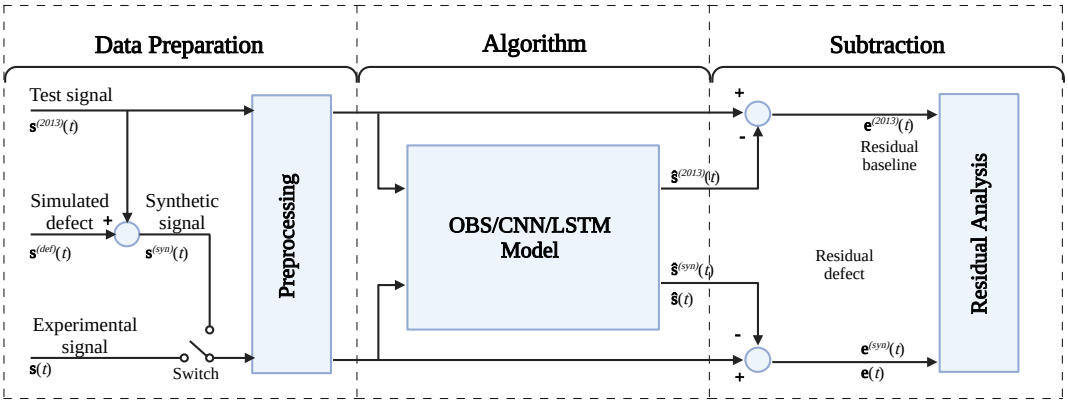


Figure 4: General residual-based defect detection process for GWSHM, adapted from Wang et al. [3]. An experimentally measured or synthetic defective signal and a known pristine signal from the structure are input into each model. The output is a residual time-trace which can then be subject to further analysis.

primary mechanisms by which temperature can alter the waveform of the recorded signals are as follows:

2.3.1 Phase velocity change

As the material heats up, its Young's modulus significantly decreases. This drop in stiffness causes a reduction in the velocity of guided waves, thus causing individual wavepacket arrival times to be delayed [14]. This phase velocity change typically causes the largest distortions to $e(t)$, as due to the nature of the algebraic subtraction, even a small time shift between two otherwise similar signals can leave large residual components in the resulting signal [2].

2.3.2 Thermal expansion

As temperature increases, the structure physically expands in response. Consequently, for a fixed sensor pair, the propagation distance travelled by the wave becomes fractionally longer, adding additional delay to the arrival time of the direct arrival wavepacket [14]. This effect is typically smaller on $e(t)$ than phase velocity change, but acts in the same direction under increasing temperature, thus contributing to the total shift in arrival time [3].

2.3.3 Transducer effects

Temperature has also been shown to have an effect on transducer bonding, causing consequent changes to the amplitude, phase and frequency content of the signals independently of their propagation through the structure. Croxford et al. have reported that these effects are significantly less than those mentioned above for small temperature variations of a few degrees [2], which encompasses the temperature range investigated in this project. It should be noted that these effects are relevant in long-term monitoring that may be subject to more extreme temperature changes and degradation to the sensing equipment.

These mechanisms are significant as they produce residual components even in the absence of structural damage. Uniform temperature changes can be approximated as a global manipulation to the entire signal, however, structures in practice can experience local temperature gradients and multiple propagation paths, which distort different wavepackets by different amounts. The baseline subtraction approach alone does not compensate for these effects [13,26]. For this reason, there is a need for more nuanced approaches with incorporated temperature compensation strategies so that residual components in $e(t)$ can be meaningfully interpreted as damage-associated features; the methods explored in this project for achieving this are overviewed in the remainder of this section, starting with OBS as the benchmark compensation method.

2.4 OBS (OD)

The primary objective of OBS is to reduce the discrepancy between $s(t)$ and $\hat{s}(t)$ caused by temperature variations [13]. This is achieved by taking an experimentally measured signal from the structure, $s(t)$, and manipulating its best-matched time-trace from the baseline library to produce an estimate for the equivalent baseline signal from the pristine structure, $\hat{s}^{(OBS)}(t)$. This is given by

$$\hat{s}^{OBS}(t) = As_k^{train}(t/\beta) \tag{2}$$

where the optimal parameters $\{A^*, k^*, \beta^*\}$ are found by minimising the MSE over a fit window, W:

$$\{A^*, k^*, \beta^*\} = \arg\min_{A,k,\beta} \sum_{t \in W} \left[s(t) - As_k^{train}(t/\beta)\right]^2 \tag{3}$$

The index k^* represents the index of the best matched time-trace in the baseline library to the inputted signal. A^* and β^* represent the optimal values of the amplitude scaling and time dilation respectively of the best matched baseline signal, where the methods used to optimise these parameters are as detailed in Section 4.1.2. It is important to note that k^* does not always correspond to a signal that was recorded at the same measured temperature as that of the input signal, and it has been shown experimentally by Chia et al. that categorising baselines by temperature yields results with a higher level of error than the OBS algorithm [13]. In the optimal case, all reflections from benign structural features are suppressed and the residual signal contains only the reflections associated with scattering from defects that appeared during the operational life of the structure. The accuracy of OBS, however, is still dependent on the extent to which the waveform distortion can be compensated by a single baseline, amplitude scale and time-dilation factor.

2.5 CNN autoencoder architecture (RJ)

To analyse defect detection with temperature compensation using guided waves, a 1D CNN was used. Guided wave traces contain localised waveform features, such as changes in amplitude, phase and waveform shape, which can be captured effectively using convolutional kernels. Convolution layers are well suited to this task because they learn local features while remaining computationally efficient through parameter sharing across the signal length. In a 1D CNN, the kernel size defines the number of adjacent samples considered by each convolutional filter at once. Convolutional autoencoders have previously been applied to unsupervised structural damage localisation using Lamb wave diagnostic signals, where they were shown to process signals without requiring prior feature extraction [27]. Defective traces were unavailable for training, so a supervised classifier could not be used. Instead, an unsupervised anomaly detection model was required. The autoencoder was trained purely using healthy traces by minimising the mean squared error (MSE) between input and reconstructed signals. As a result, defective traces are expected to produce larger reconstruction errors because their features do not align with healthy training data.

The CNN autoencoder architecture is shown in Figure 5. The initial input signal length was cropped to 1×4500 as described in 4.2. The encoder compresses the signal in three 1D convolutional layers with ReLU activation, each followed by MaxPool1D downsampling with pool size 2. Each pooling stage halves the signal length while the number of feature channels increases as shown in Figure 5. This produces a 64×562 latent representation, giving an approximate 8:1 reduction in temporal length. This was chosen to encourage the model to learn the most dominant healthy signal structure while avoiding excessive compression, which could reduce construction quality. Increasing channel depth compensates for the loss in temporal resolution as the remaining samples are defined by a richer set of learned features.

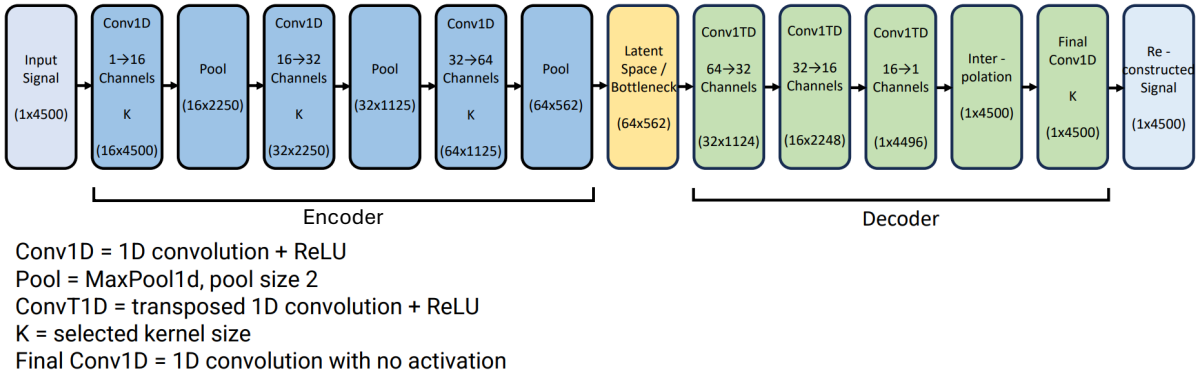


Figure 5: Architecture of the 1D CNN autoencoder used for reconstruction-based defect detection. The input signal of size 1×4500 is compressed during convolutional layers to a 64×562 latent representation before being reconstructed to the original signal length.

The compressed representation of the signal forms the latent space which acts as a bottleneck, encouraging the model to encode only the most important healthy signal features. When a defective trace is passed through the trained model, anomalous features are less likely to be reconstructed accurately, producing a larger residual error. However, the use of MaxPool1D means that small defect-related features may be suppressed during downsampling before reaching the bottleneck.

Then, the decoder mirrors the encoder using three transposed convolutional layers (ConvT1D) with stride 2, progressively restoring the signal length while reducing the channel depth. The symmetric structure provides balanced compression and reconstruction, with each downsampling stage having a corresponding upsampling stage. Due to pooling and transposed convolutional operations, the decoder does not exactly recover the initial length, so linear interpolation is applied before the final output layer to restore the signal to 1×4500. A final convolutional layer with no activation function is then applied, allowing the output to take negative values consistent with the original signal range, which a ReLU activation would incorrectly clip to zero.

The model was trained using the Adam optimiser, a ReduceLROnPlateau scheduler and early stopping, as shown in Table 7 in Appendix A.

2.6 LSTM architecture (ES)

Section 2.2 outlined the general residual generation process through baseline subtraction used by all the models in this study. This section focuses specifically on the implemented LSTM architecture, where Figure 6 summarises the overall trace reconstruction pipeline and unrolled structure of the chosen LSTM predictor. This section first describes the preprocessing and sliding-window formulation used to generate model inputs, before discussing key architectural features that make the LSTM well suited to this task.

Before input to the model, each trace was pre-processed to improve the efficiency and stability during training. The trace was first cropped between samples 500 and 5,000, corresponding to 0.1 and 1 ms, to retain the arrival window and pulse decay, so signal regions outside the area of interest were removed. Subsequently, all train, validation and test signals were standardised with equation 4,

$$\tilde{x}_{n,i} = \frac{x_{n,i} - \mu_{\text{train}}}{\sigma_{\text{train}} + \epsilon} \tag{4}$$

where μ_{train} is the mean and σ_{train} is the standard deviation, calculated from the training dataset alone. ϵ is a small constant added for numerical stability, $x_{n,i}$ is the original sample from trace n at sample index i, and $\tilde{x}_{n,i}$ is the corresponding standardised value. This provided a common input scale to the model,

whilst avoiding data leakage and preserving amplitude information between traces. Previous studies have performed a per trace peak normalisation; however, for the LSTM, this was deemed potentially too aggressive and highly dependent on the peak sample within each individual trace. Per-trace peak normalisation rescales each trace independently, making absolute amplitude and energy differences between traces less available to the model. Such pre-processing may suppress amplitude drift associated with EOCs, but in this study, it was intended for the LSTM to learn robustness to EOC-related variation.

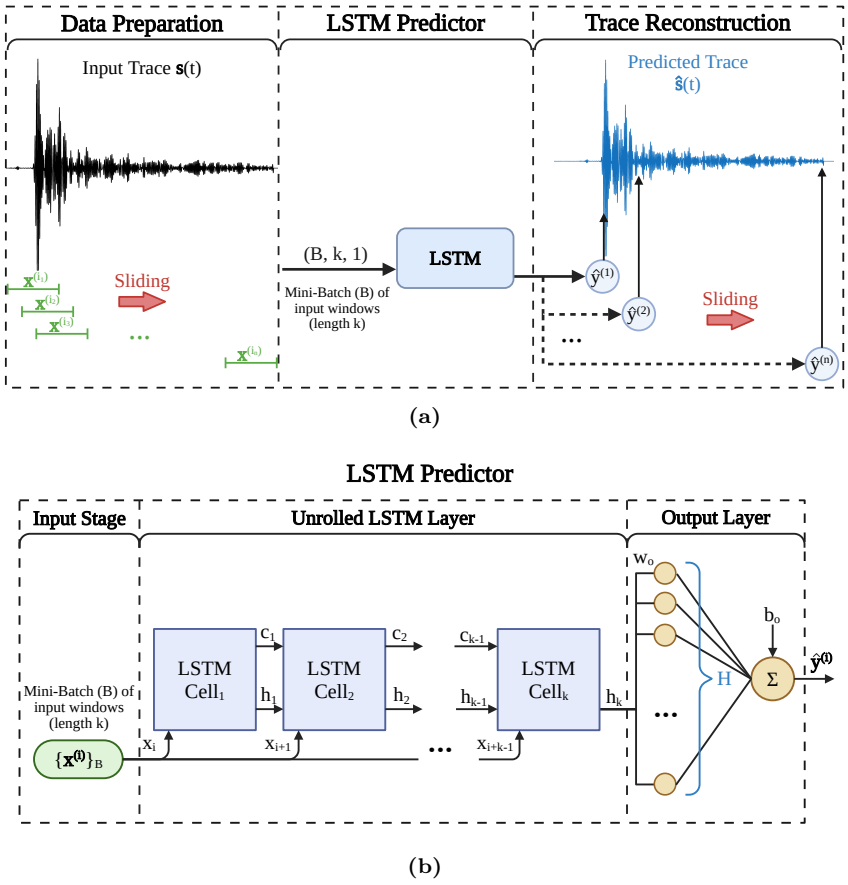


Figure 6: Schematic of the implemented LSTM trace reconstruction predictor. (a) Sliding-window strategy used to reconstruct the predicted healthy trace, $\hat{s}(t)$, from a preprocessed input trace $s(t)$. (b) Unrolled LSTM predictor for one input window, $\boldsymbol{x}^{(i)}$, where the same LSTM cell is unrolled across k time steps and the final hidden state is mapped to the output prediction, $\hat{\boldsymbol{y}}^{(i)}$.

In this study, a sliding-window methodology, similar to Wang et al. [3], was used for residual generation. Figure 6a illustrates the prediction pipeline in which a pre-processed input trace, $s(t)$, is passed through the trained LSTM to reconstruct a predicted trace, $\hat{s}(t)$. This reconstructed trace is built up from a series of individual next-sample predictions, generated using sliding history windows. Each individual history window, of length k and beginning at sample i, is extracted from the trace such that

$$\mathbf{x}^{(i)} = [x_i, x_{i+1}, \ldots, x_{i+k-1}]\,. \tag{5}$$

For single-step prediction, the corresponding prediction target is the subsequent sample,

$$y^{(i)} = x_{i+k}, \tag{6}$$

where the LSTM learns a nonlinear mapping from the input window to the next sample, such that

$$\hat{x}_{i+k} = f\left(\mathbf{x}^{(i)}\right). \tag{7}$$

By sliding this window sequentially across the trace, a set of next-sample predictions, starting at sample $k+1$, is generated and assembled to reconstruct the predicted trace, $\hat{s}(t)$, for the residual evaluation process detailed in Section 2.7.

As shown in Figure 6b, the LSTM predictor may be considered in three stages: (i) the input stage formats each history window for the network; (ii) the recurrent layer is unrolled across k time steps so that an LSTM cell processes the samples $x_i, x_{i+1}, \ldots, x_{i+k-1}$ sequentially; and (iii) the output stage maps the final hidden state to the predicted subsequent sample, $\hat{y}^{(i)}$, through a fully connected layer. The input to the LSTM is arranged as a three-dimensional tensor of shape $(B, k, 1)$, where B is the batch size, k is the history window length, and the final dimension represents the single input feature for each time step. Each batch therefore contains B sliding windows, with each window formed from k consecutive amplitude samples extracted from the guided-wave trace. This history length, k, defines the number of previous samples exposed to the model for each next-step prediction and corresponds to the number of recurrent time steps over which the LSTM is unrolled. The input size is 1 because the signal amplitude is used as the only predictive feature, rather than also including phase information.

Within the unrolled recurrent layer, the k samples of each input window are processed sequentially by the same LSTM cell, allowing temporal information to propagate across the full history window. The amount of information that is retained and passed through the hidden and cell states, acting as the LSTM memory pathways, is regulated through the LSTM's internal gated structure, which is well established in previous literature [28] [29] [30]. Within LSTM cells, the hidden size, H, is a tuneable parameter that defines the size of the internal feature representation carried through the recurrent layer.

Having processed the entire history window, a final hidden state representation, h_k, carrying the learned temporal information is passed to the output layer. Within this output layer, a final weight and bias are applied to the H-dimensional feature representation and a fully connected layer maps this vector to a single next-sample prediction. All the next-sample predictions, $\hat{y}^{(i)}$, are then used for trace reconstruction as previously described.

The described LSTM architecture is well suited to this task of waveform prediction. Through its gated structure, the LSTM can learn which information should be retained, updated, or discarded, providing a more adaptive temporal representation than a fixed explicit lag-based formulation used by the NARX. This architectural difference is what justifies the hypothesis described in Section 1. LSTMs are also designed to mitigate the vanishing-gradient problem commonly experienced by simpler RNNs, making the selected model suitable for modelling waveform dependencies across the input history window.

2.7 Evaluation metrics (ES)

To assess the quality and sensitivity of the three residual-generation-based approaches, a common criterion is required to directly compare how effective each model is for defect detection. The metrics used in this project are standard and are detailed in previous literature, including Wang et al. [3]. For this reason, this section does not exhaustively explain receiver operating characteristic (ROC) curves or the calculated area under the curve (AUC), and instead provides a summary of these metrics.

ROC curves are computed using the probability of detection (POD) and probability of false alarm (PFA). A threshold sweep is performed over the set of healthy and defective residuals, where the peak residual amplitude is used for defect classification. If the peak residual amplitude is greater than the threshold, the trace is classified as defective; otherwise, it is classified as healthy. For healthy residuals, this counts as a false alarm, while for defective residuals, it counts as a successful detection. A perfect detector is one

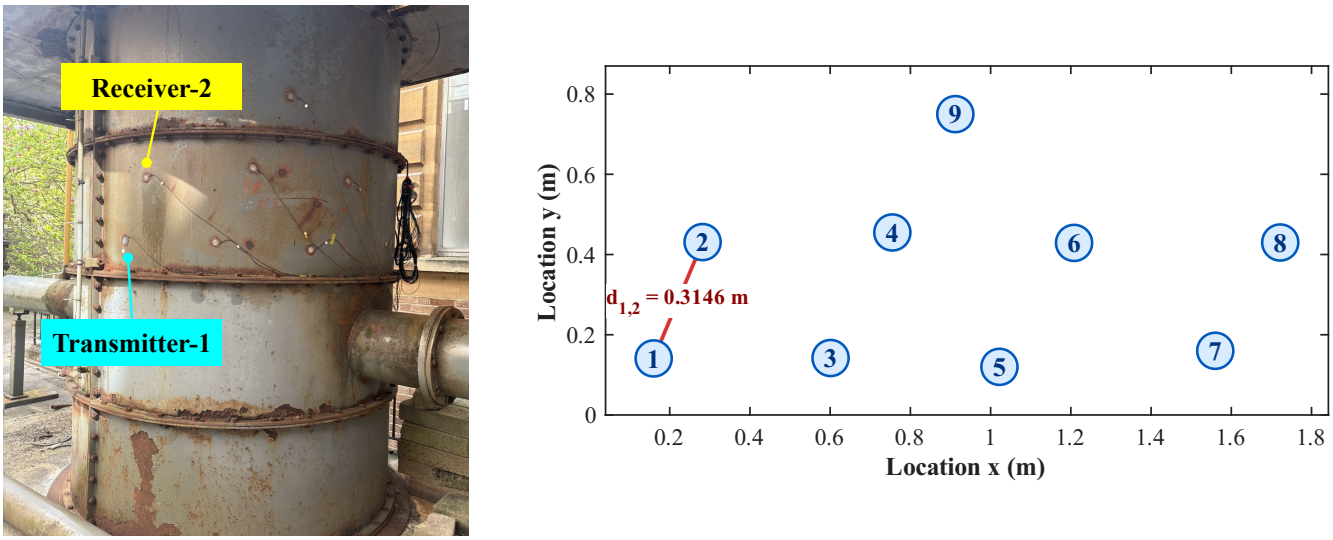
for which a threshold exists that correctly classifies all residuals. AUC is computed as the area under the ROC curve, with a perfect detector having an AUC of 1.0 and a random classifier having an AUC of 0.5.

In this study, two threshold-sweep strategies are used. For OBS, a strategy matching that used for the NARX model in Wang et al. [3] was applied, where an arrival window of 0.05 ms around the known defect arrival time was used to evaluate the residual. This approach reduces the effect of random residual spikes outside the defect arrival window, making the thresholding strategy more robust to non-defect-related discrepancies in the residual. The CNN and LSTM use a different approach, where the defect arrival time is assumed not to be known; this is a more common situation in industry and in practical GWSHM. This threshold sweep is taken across the whole residual trace, resulting in a detection process that is more sensitive to random spikes outside the defect arrival region. Whilst this strategy requires a more globally even and lower-amplitude healthy residual, it provides a more realistic scenario for defect detection in industry.

3 Data Collection (OD)

3.1 Set-up of water tank (OD)

The performance comparison of the OBS, CNN and LSTM models for damage detection is completed using data from a single sensor pair, Tx1-Rx2, from a sparse array of nine sensors on a water tank located at the University of Bristol. Figure 7 shows the tank and the reconstructed sensor array with the distance between the two sensors of interest, $d_{1,2}$.



(a) (b)

Figure 7: The sensor pair Tx1-Rx2 on (a) a photograph of the tank and on (b) the reconstructed sensor array showing the locations of all sensors, with $d_{1,2}$ identified as 0.3146 m.

Primary data collection was not carried out as part of this project; instead, the tank data was acquired from the existing experimental set-up described by Wang et al. [3], as this dataset captures long-term EOC variation over a period much longer than could be achieved within the timescale of this project. The experimental parameters from this study are reproduced in the remainder of this subsection.

3.1.1 Structural setup

The tank is approximately 2 m in diameter and fabricated from 5-mm-thick, semi-cylindrical steel panels that have been bolted together along flanges. The sealant used between the flanges is unknown. The tank

remained empty throughout the data collection period, and is located in a sheltered exterior environment prone to natural environmental weather and temperature cycles.

3.1.2 Sensor configuration

A sparse array of permanently attached piezoelectric disks (NCE51, ∅20 mm × 1 mm thick) are bonded to the tank with cyanoacrylate adhesive and protected by a silicone sealant. The system is designed to excite and detect the S_0 guided wave mode at a center frequency of 250 kHz. At 250 kHz in 5-mm steel plates, the S_0 mode is the fastest fundamental mode and shows minimal dispersion [3].

3.1.3 Signal generation and processing

Due to the hardware output being limited to 12 V maximum, a 1000-cycle chirp excitation signal designed with frequency content mathematically equivalent to a 5-cycle Hanning-windowed toneburst centred at 250 kHz is used to increase the transmitted energy without requiring a larger voltage. This is then filtered in post-processing using the ratio of the desired toneburst to the transmitted chirp in order to recover the equivalent narrowband toneburst response in the tone domain for residual analysis, which retains the high signal-to-noise (SNR) advantage of the longer chirp excitation [31].

3.2 Discussion of dataset trends (OD)

The data from sensor pair Tx1-Rx2 used in this project spans approximately 19.5 months. A summary of how the dataset was split and the trace counts are shown in Table 1. $\boldsymbol{s}^{(2012)}$ is used as the training dataset, in which the signals are assumed to be from the pristine tank under a full calendar year of varying EOCs. $\boldsymbol{s}^{(2013)}$ forms the basis for the testing dataset. These signals are assumed to come from a near-pristine structure, and thus are not sufficient alone to quantify the performance of the three models for defect identification. To combat this, they are used as the basis of the synthetic defect signals created to assess detection performance.

Table 1: Summary of the Tx1-Rx2 dataset and how it has been used in this analysis.

Dataset	**Date range**	**Number of traces**	**Role in analysis**
Full dataset	4 Jan 2012, 09:28 to 20 Aug 2013, 14:00	32,695	Long-term dataset containing naturally varying EOCs
$s^{(2012)}$	2012 monitoring period	18,455	Healthy ML training dataset and OBS baseline pool
$s^{(2013)}$	2013 monitoring period up to 20 Aug 2013	14,240	Near-pristine test dataset for synthetic defect generation

Across the range of the data, clear variations to the baseline properties of the signals due to the naturally occurring EOCs can be observed. These are shown visually in Figure 8, which illustrates how both the RMS amplitude and the estimated group velocity change with time.

3.2.1 RMS

The raw RMS in Figure 8(a) shows the daily fluctuations between day and night, whilst the dashed daily median line highlights the macro-level seasonal changes. A noticeable dip in the RMS can be observed in the winter months, whilst the peaks occur during summer. On top of the seasonal cycles, a general long-term upward/downward drift can be seen, potentially indicative of deterioration of the tank or its sensors and their bonding. This plot serves to demonstrate that the dataset signals change over time in the absence of a structural defect, and validates its use for analysis as a long-term GWSHM scenario.

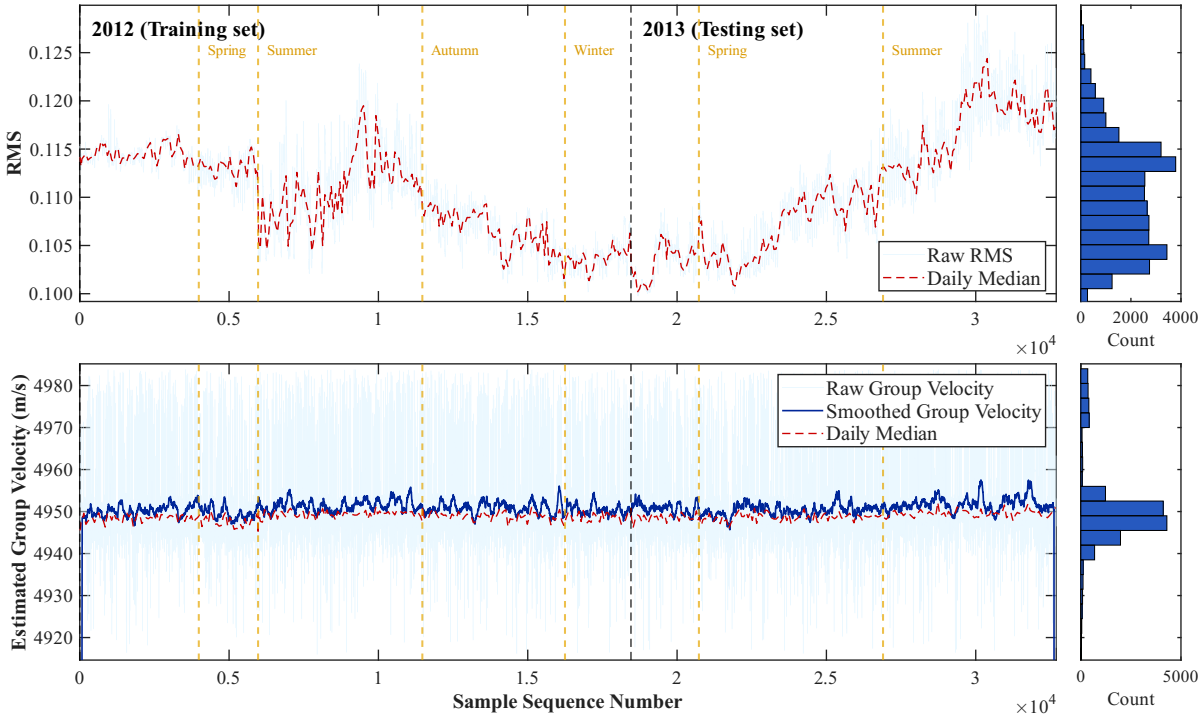


Figure 8: (a) The RMS amplitude and (b) the estimated group velocity of the Tx1-Rx1 signal path. The sample sequence number represents the chronological measurement order. The light blue lines show raw trace-to-trace variation, and the dashed red lines show longer-term trends. The vertical yellow dashed lines separate the four seasons across the calendar year.

3.2.2 Group velocity

The group velocity, v_g, in Figure 8(b) is estimated using the time of first arrival in the baseline signal envelope and the fixed distance between the two sensors. The velocity distribution is 4948.9 ± 10.3 m/s, which is in an agreement range of 2.8% - 3.8% (including the stated uncertainties) of the theoretical value of 5100 m/s for the group velocity of S_0 lamb waves in a steel plate with frequency-thickness product 1.25 MHz·mm [32], which is that of the tank.

The light blue raw group velocity demonstrates the sensitivity to immediate temperature fluctuations, whilst the darker blue and dashed red lines show that the underlying wave speed is relatively stable. The sensitivity of an S_0 wave to temperature is the same of that of a bulk longitudinal wave to a first approximation. From textbook values for the sensitivity of Young's modulus of steel to temperature, this is estimated to be approximately 1.3 $\text{ms}^{-1}/°\text{C}$ [3]. The average temperature change across the year of 2012 in Bristol was approximately 18°C [33], which equates to an expected wave speed shift of approximately 23.4 m/s across the year. This is consistent with the spread of 20.6 m/s across the dataset. This validates that the velocity shifts in the data are caused by changes in the outdoor temperature, and reinforces the critical need for temperature compensation strategies to counteract this effect in the dataset.

3.3 Defect simulation and synthetic defect dataset (OD)

In order to produce quantifiable detection performance comparisons between the three models, a large range of data with known defects is required. This is not feasible to obtain experimentally for the tank due to the amount of time since the initial data collection took place - the tank and its sensors have now deteriorated to the point of being beyond use. Additionally, producing an adequate dataset would require introducing defects of known severity at a variety of different locations under a representative range of EOCs, which is impractical to obtain experimentally. As a result, simulated defect responses, $s^{(def)}(t)$, were linearly superposed onto 200 randomly sampled experimental signals from s^{2013}, given by

$$s^{(syn)}(t) = s^{(2013)}(t) + s^{(def)}(t) \tag{8}$$

where $s^{(syn)}(t)$ is the resulting synthetic defect signal. This creates defective data that preserves the naturally occurring EOC variations from the tank data whilst allowing defect parameters to be controlled.

3.3.1 Modelling

The simulated defect response is modelled as a delayed, scaled copy of the 5-cycle Hanning-windowed 250 kHz toneburst signal transmitted by the experimental setup. The method used for this closely follows that as carried out by Wang et al [3], and the full mathematical derivation of the amplitude scaling factor can be found in their $\alpha = 1$ case. This is representative of the case in which the arrival time of the defect (and therefore the total propagation distance) is known, but the distances from the defect to each of the transmitter and receiver are unknown. The defect must lie on an ellipse with the transmitter and receiver at the foci [3]; a conservative assumption is thus made that the position of the defect on the ellipse is that corresponding to the minimum defect response amplitude due to beam spread. The full simulated defect response, $s^{(def)}(t)$, for this case is represented by

$$s^{(def)}(t) = A_{1,2}\sqrt{d_{1,2}}\left(\frac{2}{c \cdot t_d}10^{\beta/20}\right)s^{(ref)}(t - t_d) \tag{9}$$

where $A_{1,2}$ is the amplitude of the directly transmitted signal, β is the decibel (dB) scaling factor and $s^{(ref)}$ is the shape of the transmitted signal with unit amplitude and zero delay. The defect arrival time, t_d, is parameterised with the detection range, τ_r. This is the delay of the defect response relative to the first direct arrival of the transmitted signal, given by

$$\tau_r = t_d - d_{1,2}/c \tag{10}$$

where t_d is the defect response arrival time. For each selected value of τ_r, the simulated defect response is centered at the corresponding t_d. For the comparison in this project, the detection range is incrementally increased over $\tau_r \in [0.12, 0.5]$ ms. This corresponds to 20 absolute arrival times from 0.1836 ms to 0.5636 ms, which all occur after the first direct arrival wavepacket in the time-trace. Five defect amplitudes, β = -10, -15, -20, -25 and -30 dB, are selected for the analysis. For each combination of defect amplitude and arrival time, the resultant simulated defect is added to the prior sampled 200 pristine s^{2013} signals. This forms a controlled synthetic dataset of known defect amplitudes and arrival times, with realistic structural response and environmental variability from the tank. An example trace is shown in Figure 9.

3.3.2 Physical simplifications

Any secondary effects, such as shadowing of structural reflections, were ignored in this model due to the difficulty in replicating exact geometry-dependent boundary conditions. It should thus be noted that the purpose of the synthetic data is simply to provide a detection sensitivity benchmark through use of a controlled response with known parameters, and cannot be used to indicate real operational performance. The level of confidence that can be drawn from the comparisons of this data are thus governed by the assumptions in Table 2.

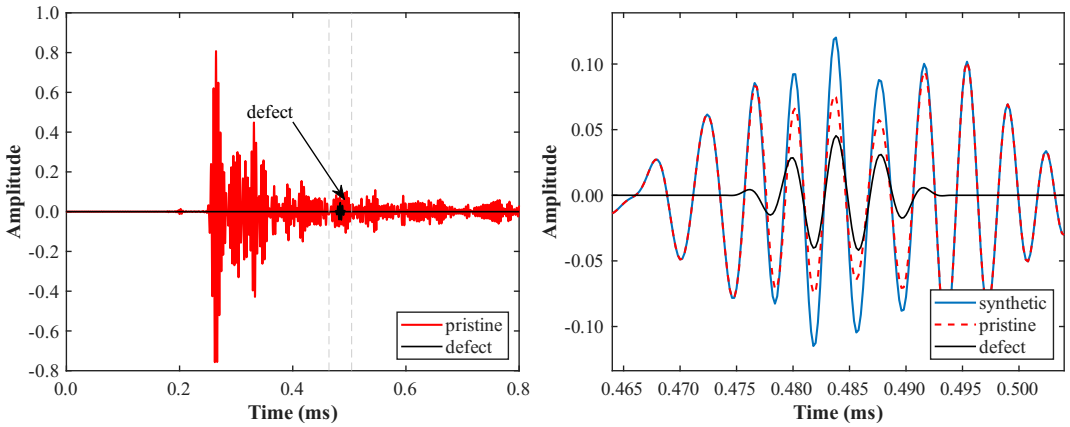


Figure 9: Synthetic defect trace formed by superposition of a simulated defect onto a pristine signal from s^{2013}. The defect has amplitude -25 dB and arrival time 0.484 ms. Its location is shown on (a) the full pristine trace and (b) a zoomed in plot of the injected defect region.

Table 2: Summary of the assumptions made in the synthetic defect dataset and their effect on the subsequent analysis.

Assumption	Motivation	Effect on analysis
Defect response is a delayed and scaled toneburst	Enables controlled adjustment of defect amplitude and arrival time	Simplifies the physics of real defect responses
Linear superposition is used to integrate simulated defect	Preserves measured pristine tank response whilst incorporating known defect response	Does not account for how damage alters other parts of the waveform
Same 200 pristine traces are used across conditions	Enables fair comparison across all amplitudes and arrival times	Results dependent on how representative these traces are of the full EOC distribution in s^{2013}
Arrival time is controlled	One pitch-catch arrival time corresponds to an ellipse of possible locations	Dataset only able to evaluate detection sensitivity, not localisation accuracy
Shadowing is ignored	Maintains consistent, controlled data	Real defects may alter or reduce later structural reflections [34]
Mode conversion is ignored	Avoids detailed physical scattering model for the tank	Real defects could generate additional modes not accounted for [35]

4 Numerical Methods (OD, RJ, ES)

4.1 OBS (OD)

This section details how the OBS model was constructed and implemented to act as a comparison benchmark against the two ML models, alongside a discussion of the main limiting assumptions. A visual schematic for the full OBS workflow is shown in Figure 10.

4.1.1 Data preparation

The OBS method compiles a baseline library $\mathbf{s}^{(train)} = \{s_{k_{train}}(t)\}$, where $k_{train} = 1 \dots N_{train}$. For this analysis, $N_{train} = 20$ signals were uniformly sampled from $\boldsymbol{s}^{(2012)}$. This subset satisfies two critical requirements governing the accuracy of OBS: (i) the full seasonal range of EOCs experienced by the structure whilst deemed pristine is covered and (ii) the long-term reduction in signal amplitude caused by the natural deterioration of the tank and its sensors is embedded into the library.

All signals, including the input signal, are then amplitude normalised by the maximum of their analytic envelope. In pitch-catch configuration, the first dominant wavepacket in the time-trace typically cor-

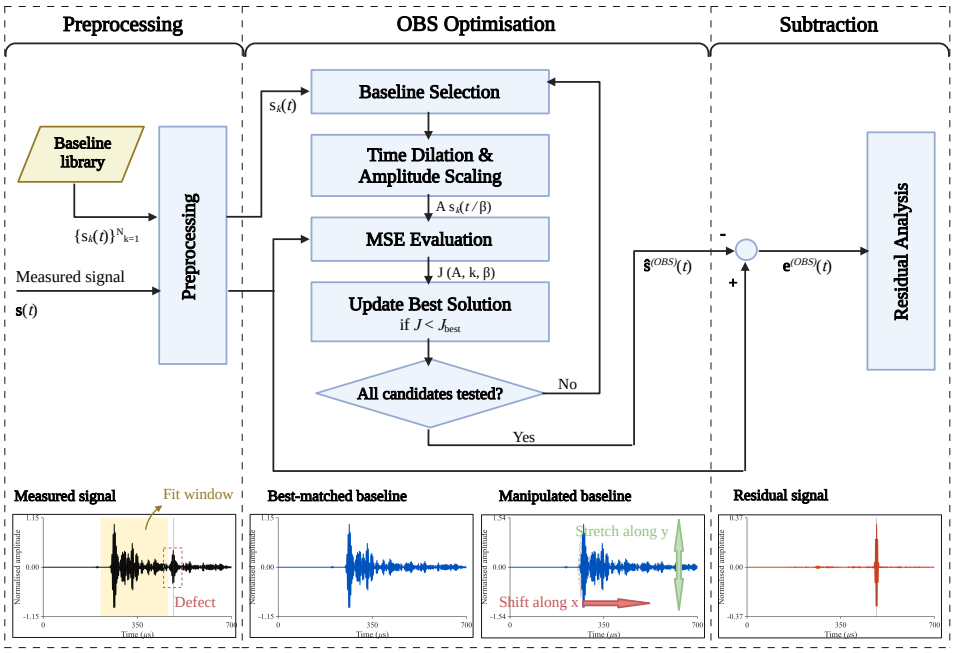


Figure 10: Schematic of the OBS algorithm. The measured signal, s(t), is preprocessed then compared with all candidate baselines in the baseline library. Optimal time dilation and amplitude scaling parameters are calculated by minimising MSE over a fit window and the resultant equivalent pristine signal is subtracted from s(t) to produce the residual. The manipulation of the signal at each stage of the process can be seen by the flow at the bottom of the schematic.

responds to the direct arrival [2], giving a phase independent measure of signal energy. This serves to eliminate the effect of fluctuations in the signal due to non-structural variations, such as instrumentation drift and adhesive degradation. This ensures that the optimisation procedure is constrained only to the local shifts and scattering associated with structural defects rather than global fluctuations in the signals.

4.1.2 OBS fitting and residual formation

Croxford et al [2] numerically approximate temperature-induced velocity changes of guided waves as fractional stretches/compressions of the time axis, and propose the use of a time dilation factor to compensate. Following this, the input signal $s(t)$ is matched against every candidate baseline in $s^{(train)}$ and for each one a global time dilation factor, β, is applied. The candidate dilated baseline, $s_{k_{\text{train}},\beta}(t)$, is given by

$$s_{k_{\text{train}},\beta}(t) = s_{k_{\text{train}}}\left(\frac{t}{\beta}\right) \tag{11}$$

The scaling of t/β does not typically coincide with the original sampling points of the experimentally captured baselines, and so temporal interpolation is used to evaluate the dilated signal.

For each candidate pairing of baseline and time dilation factor, an amplitude scaling factor is calculated over the localised fit window, W, using ordinary least squares:

$$A = \frac{\sum_{t\in W} s(t)s_{k_{\text{train}},\beta}(t)}{\sum_{t\in W} s_{k_{\text{train}},\beta}(t)^2} \tag{12}$$

This step compensates for any remaining global differences in amplitude after the envelope normalisation. W is deliberately placed around the direct arrival wavepacket as this region is high amplitude and phys-

ically corresponds to the direct transmitter-receiver path. The optimal parameters are then finalised by minimising the MSE over W, given by

$$J(k_{\text{train}}, \beta) = \frac{1}{N_W} \sum_{t \in W} \left[s(t) - A(k_{\text{train}}, \beta) s_{k_{\text{train}}, \beta}(t) \right]^2 \tag{13}$$

where N_W is the number of discrete samples. The combination giving the minimum J is the optimal OBS fit. The equivalent pristine baseline is then constructed over the full length of the signal and subtracted from the input using radio frequency (RF) subtraction to produce the residual $e^{OBS}(t)$:

$$e^{OBS}(t) = s(t) - \hat{s}^{OBS}(t) = s(t) - A^* s_{k^*, \beta^*}(t) \tag{14}$$

Croxford et al [36] compared signal subtraction on the signal's envelope against the RF waveform for sensors operating in pitch-catch mode, suggesting that RF subtraction is the superior method; since phase information is lost when the envelope subtraction is calculated, the remaining signal shows complex interference which creates a series of "blind spots". The final values of the parameters used for OBS fitting are shown in Table 3.

Table 3: Final OBS settings used for the analysis conducted in this project.

Quantity	**Value**	**Justification**
Baseline library size	$N_{\text{train}} = 20$	Selected from sensitivity study as a performance/runtime compromise whilst maintaining EOC variation
OBS fit window, W	[−0.05, +0.30] ms (around direct arrival)	Captures high amplitude direct wavepacket while avoiding excessive later reflections
Dilation range	$\beta \in [0.95, 1.05]$	Sufficient to cover temperature range expected to be endured in typical operational life [2]
Number of β values	401	Provides fine grid search resolution

4.1.3 Sensitivity studies

To determine the final selected values of N_{train} and W in Table 3, two sensitivity studies were conducted with the objective of achieving reliable detection performance whilst maintaining low computational cost.

4.1.3.1 Baseline library size study

To assess the variation of N_{train}, W was kept constant at [-0.05, +0.30], which was chosen by eye to cover the direct arrival wavepacket. The chosen evaluation metrics were the median global RMS to measure the typical residual floor and the 95th percentile peak residual to quantify false alarms. Random baseline selection was repeated 20 times and the median taken, which was then compared with evenly spaced baseline selection in Figure 11.

It can be observed that increasing N_{train} from 2 to 20 significantly improved the OBS algorithm with the median global residual RMS decreasing from 0.029 to 0.020 (32% reduction) and the 95th percentile peak residual decreasing from 0.246 to 0.183 (26% reduction). Beyond this, further increases to N_{train} yielded minimal improvements to performance across all metrics with a larger comparative computational cost. $N_{train} = 20$ was chosen as a practical compromise between runtime and robust EOC coverage.

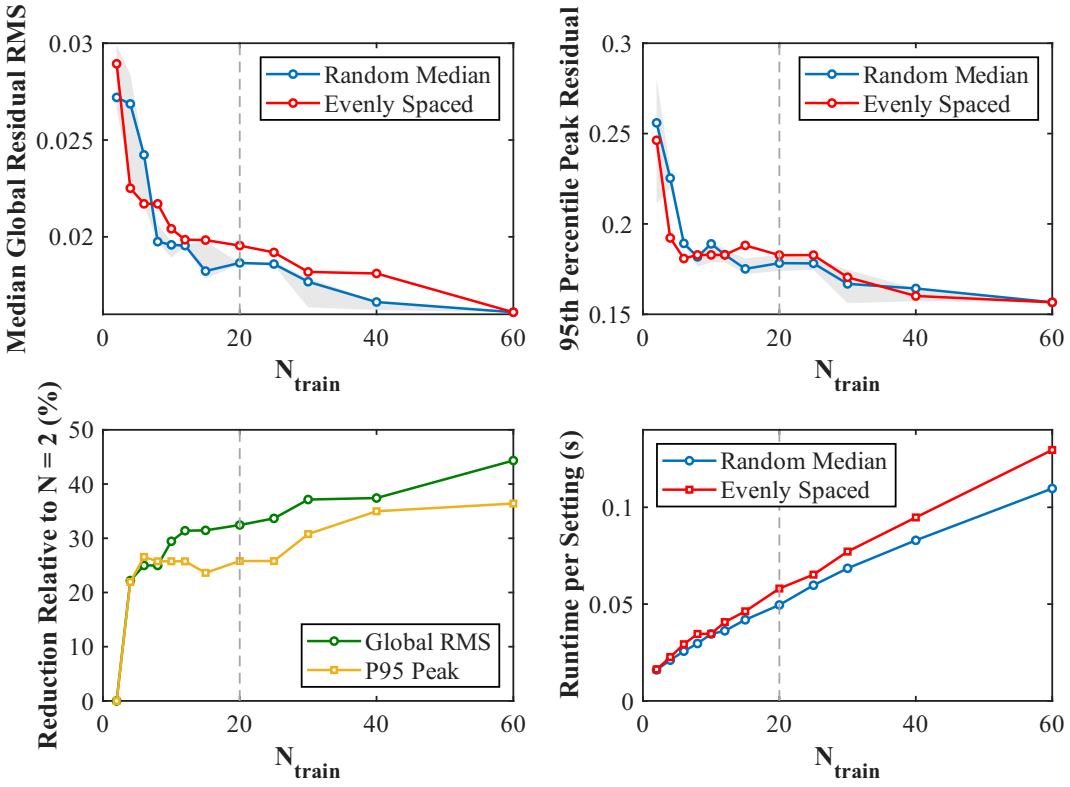


Figure 11: N_{train} sensitivity studies showing the effect of the size of the baseline library on (a) median global residual RMS, (b) false alarm peak residuals, (c) % reduction relative to the smallest tested value of N_{train} and (d) the computational cost. The dashed grey line shows the selected value of $N_{train} = 20$.

4.1.3.2 Fit window length study

The fit window size, W, was then evaluated to see if any deviation from [-0.05, +0.30] yielded better results. The pre-arrival window was fixed at −0.05 ms and the post-arrival window was then varied, whilst all other parameters remained constant. OBS was then run on two weaker defect amplitudes, -20 and -25 dB, at six evenly spaced arrival times spanning [0, 0.5] ms. Figure 12 shows the results against two comparison metrics.

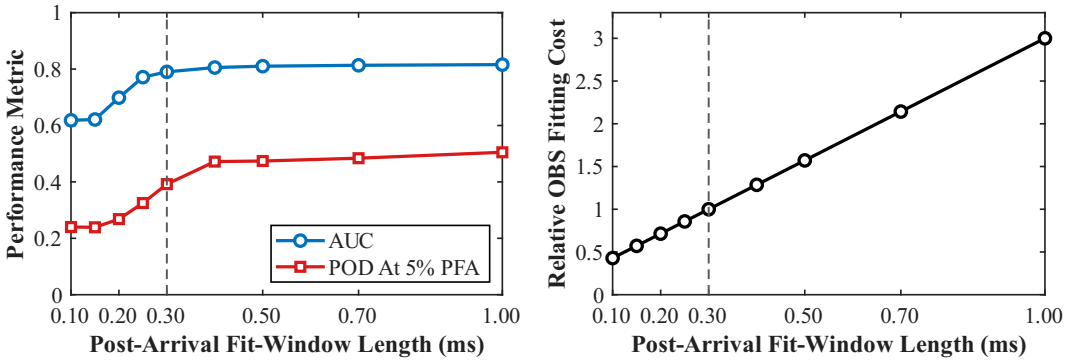


Figure 12: Plots showing the effect of varying post-arrival fit window length on (a) AUC and POD at 5% PFA and (b) computational cost, which was estimated from the number of samples in the length of fit window.

Figure 12(a) shows that performance for both metrics improved as the post-arrival window length was increased from +0.10 ms to +0.30 ms. The selected window produced an AUC of 0.790, which was only a 3.14% deviation from the best AUC in the plot of 0.816. The POD at 5% PFA for the chosen

window was within 22.4% for the best scoring window length, however, the absolute difference was small with an increase of 0.113. Despite the continued increase of POD beyond this, the rate of improvement became more gradual whilst the computational cost increased linearly. [-0.05, +0.30] ms was chosen as the best compromise to maintain the benefits of improved POD and AUC whilst keeping computational cost manageable as opposed to selecting the absolute optimal performance.

It is important to highlight that the sensitivity studies only validate the chosen settings for the available 2012-13 data. Wang et al. demonstrated that OBS performance, for a different sensor pair in the same experimental set-up, had degraded to the point of being useless towards the later stages of their 8-year study [3]. This suggests that sampling $N_{train} = 20$ from $\boldsymbol{s}^{(2012)}$ is fundamentally insufficient to capture long-term effects on the tank, but since the test period detailed here only spans 7.5 months, it is not something that could be predicted from analysing $s^{(2013)}$ alone.

4.1.4 Assumptions and limitations

OBS has clear benefits as a model; it is simple in comparison to data-driven techniques and only requires pristine baseline data to function. Despite this, a number of assumptions are being made which inherently limit its performance, which have been detailed in Table 4.

Table 4: Assumptions made by the chosen OBS model ranked by severity, alongside their limitation on the analysis. Potential mitigation strategies (including state-of-the-art contributions overcoming some of the limitations) are shown that could be used given further time and resources for a more comprehensive model.

Assumption	Severity	Limitation	Possible Mitigation
EOC effects can be represented by global amplitude scaling and time dilation	High	Different wavepackets can experience different local amplitude changes and time shifts [3]	Local phase matching or path specific compensation [37]
The baseline library sampled from s^{2012} spans future EOCs	High	Structural ageing, sensor drift, boundary changes and climate variation may move the structure outside the baseline range	Adaptive baseline updating or scalable temperature compensation methods [7]
Temperature variation is approximately homogeneous	Medium	Actual structure can have local temperature gradients affecting different paths differently	Location specific temperature compensation [38]
Scattering from structural features is independent of temperature	Medium	Boundary reflections may change with EOCs and compromise validity of residual interpretation	Physics-informed compensation of phase and amplitude [39]
Single dominant mode, S_0, is assumed	Medium	Real signals may include mode conversion or multimodal content, complicating residual interpretation	Mode filtering or time-frequency methods [20, 35]
Parameter studies generalise beyond this dataset through use of a single sensor pair	Low	Optimal N_{train}, fit window and dilation range may change for other paths or structures	Repeat parameter studies across more sensor pairs, structures and EOC ranges

4.2 CNN Hyperparameter Study (RJ)

Before any relevant hyperparameter studies, the data needed to be pre-processed so that it was suitable for the CNN autoencoder. The traces were normalised using the same method as the OBS, as described in Section 4.1.1. Despite being complex guided-wave traces, only the real component was used. This was justified because the dominant defect-related changes were expected to remain visible in just the real component and reduced computational complexity by reducing the input to a single channel. Since the objective of the study was to determine that the CNN autoencoder could distinguish between reconstructed healthy and defective traces by finding the reconstruction residuals, it was considered reasonable to begin with the reduced signal representation before adding the complexity of the full complex traces.

This provided an input shape of (N, C, L), where N is the number of traces, C is the number of channels and L is the length of each trace.

The traces were then cropped from an initial length of 6000 samples to a length of 4500, removing the first 500 and last 1000 samples. This was done so that the regions of the signal with little data were removed so the model could focus on the section of the waveform where defects would most likely show. For each model a training set size of 1500 traces and a validation set size of 500 was chosen. This allowed for a reliable assessment of the model's performance while keeping the training and validation process computationally manageable. The training and validation set was taken from a range of dates within 2012, allowing for variety of healthy trace shapes to be considered during model development. The test set consisted of 200 corresponding healthy and simulated defective signals from 2013. Data from a different year was used so that the model was evaluated on traces that were not taken from the same period as the training and validation data [30]. This tested whether or not the model had learned general healthy signal behaviour rather than fitting to the specific conditions present in the 2012 data.

4.2.1 Kernel study

The main study investigated how changing the kernel size affected the residuals produced. Kernel size controls the temporal receptive field of the CNN autoencoder, changing how much of the guided-wave trace each convolutional filter considers when moving along the trace to extract waveform features [40–42]. For a small kernel size, only short sections of the signal are considered, allowing localised features to be learned, but may struggle with wider wave packets and longer oscillatory structures. Larger kernels learn broader guided-wave features more easily because each filter covers a longer section of the trace. However, increasing the kernel size increases the number of learnable parameters, which can make the model more prone to learning trace specific variations or noise in the training data rather than general waveform structure [43].

Another part of the CNN that changes with kernel size is the padding. Padding was added so that the convolution does not shrink the feature-map length at each convolution stage. Increasing the kernel size reduces the output length because the kernel cannot be centred at the signal boundaries without extending beyond the available samples [44]. It was important to ensure that the signal is not shortened at each convolution so that the boundary regions of the signal could still contribute to the convolutional output. Zero padding was used, which adds zeros to the end of the signal. Although this does not add new physical information to the signal, it allows for consistent boundary treatment and prevents differences in output length for models with different kernel sizes [44]. For a CNN with a stride of one, padding is calculated using Equation 15 as shown below:

$$p = \frac{k-1}{2} \tag{15}$$

where p is the padding applied to each side of the signal and k is the kernel size.

This allows for the same number of samples to be added to each end of the signal, maintaining the feature-map length through each convolution and therefore making the CNN architecture easier to control. The benefit of using this method for the study was that it allowed different kernel sizes to be compared while keeping the feature-map dimensions constant, meaning that changes in model performance were due to the kernel size, not due to changes in output signal length. This meant that kernel sizes tested had to be odd to give an integer value for padding.

For the kernel size study, the architecture and training settings remained constant except for the kernel size. The same clean dataset was used for training and validating each model. During testing, the same pairs of clean and simulated defective data were used on the model, with the reconstructive error used

to produce a residual [45]. This ensured that differences in performance could primarily be attributed to changes in kernel size rather than changes in training set up or data selection. Initially, a broad sweep of kernel sizes between 5 and 159 were tested to identify the region that would produce stronger performance. Validation MSE was used as an early indicator of reconstruction quality, allowing different kernel sizes in those regions to be trained. A lower validation loss suggests that the model has learned the signal structure more accurately. However, since defect detection was the aim of the model, validation MSE cannot be used as the only performance measure [45]. A model that produces a low validation MSE will do so since it reconstructs a healthy signal very well. However, it may also reconstruct a defective signal with similar accuracy, making the model insufficient for anomaly detection due to reduced contrast in reconstruction errors. Therefore, validation MSE was only used as an initial baseline for comparison of kernel sizes to create a shortlist for further testing. After training, the validation MSE values for the different kernel sizes were recorded and plotted, as shown in Figure 13a.

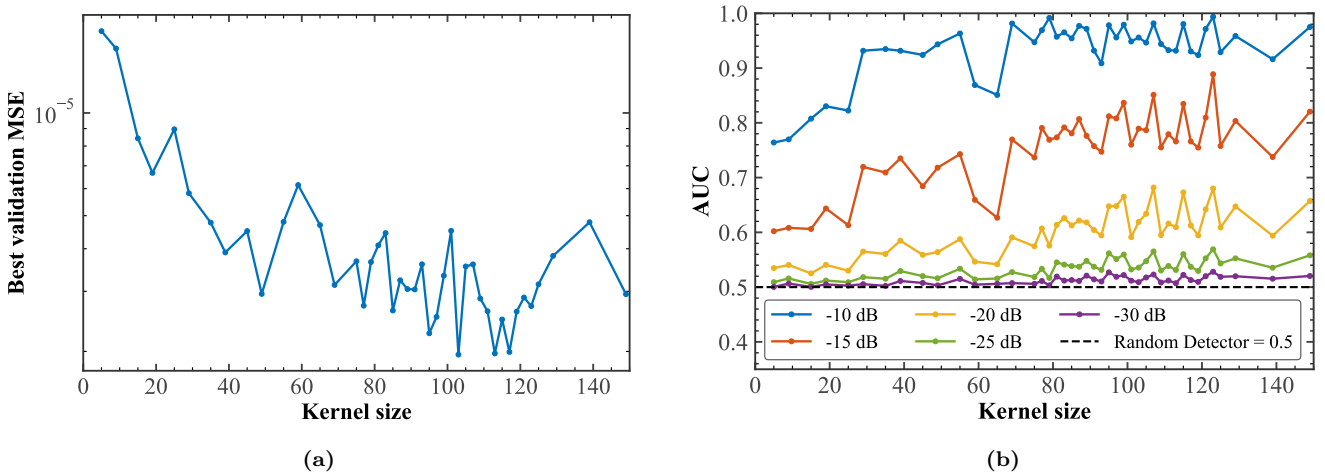


Figure 13: Influence of convolutional kernel size on CNN autoencoder performance where (a) is the best validation MSE obtained during training for each kernel size, indicating the model's ability to reconstruct healthy validation traces, and (b) is the AUC values for the same kernel sizes, evaluated at a defect arrival time of 0.344 ms with separate curves showing each simulated defect amplitude.

The results show a general trend between validation MSE and kernel size. Small kernel sizes produced higher validation errors, showing that they were less effective at capturing the relevant waveform structure. As kernel size increased, the validation MSE generally decreased with the model reconstructing traces more accurately. Several neighbouring kernel sizes produce noticeably different results, indicating that other factors influenced model performance other than kernel size. The noisy behaviour may be partly due to the stochastic nature of neural network training, such as random weight initialisation, which affects the optimisation trajectory [46]. Early stopping may also contribute to this if the training terminates before the model fully converges [47]. Once the kernel size is sufficiently large to observe the required waveform shapes, the influence of kernel size becomes less dominant compared with training variability. For future improvement on the model, the early stopping counter could be increased to ensure complete training, or repeated training of each kernel size to establish the mean and standard deviation of the best validation MSE to be calculated [46]. This would reduce the influence of anomalous individual runs and provide a more reliable comparison between kernel sizes.

ROC curves were also used to determine how well the reconstruction error separated healthy and defective signals across different detection thresholds. The area under the curve (AUC) value was used as a comparison metric [48]. This was a more suitable metric for defect detection than validation MSE because it measured how effectively the reconstruction error differentiates healthy and defective traces, rather than only measuring how well healthy validation signals were reconstructed [21]. Therefore, AUC aligned more closely with the objective of the model as an anomaly detector. Additionally, it allowed for better

comparison with the other models researched in the project. The AUC results for the different kernel sizes are shown in Figure 13b.

These results show that all kernel sizes perform poorly for weaker defects at −25 dB and −30 dB. The general trend is that larger kernels produce improved detection performance. Future testing could therefore investigate kernel sizes beyond 149, as the AUC had not clearly decreased by that point. The best performing kernel was 123, giving the highest AUC for each defect size. This shows that the previous validation MSE metric may not be the most suitable, as the same kernel size performed averagely in comparison to other kernel sizes in the same region. The relationship between validation MSE and mean AUC is shown in Figure 14, which further reinforces that low validation MSE does not necessarily correspond to stronger defect detection.

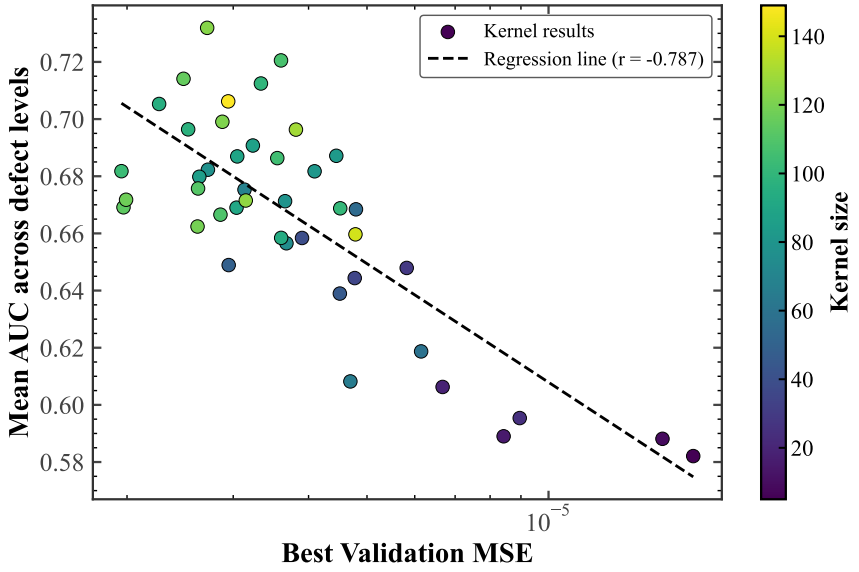


Figure 14: Relationship between validation MSE and mean AUC across all defect amplitudes at a defect arrival time of 0.344 ms. Each point represents a trained CNN autoencoder with a different kernel size, which are colour-coded according to their kernel size.

Figure 14 shows a moderate negative correlation ($r = -0.787$) between validation MSE and mean AUC across defect levels at the 0.344 ms arrival time. This indicates that a lower reconstruction error generally correspond with improved anomaly detection performance [48]. However, there is still a considerable scatter around the regression line, indicating that validation MSE alone is insufficient to predict detection performance reliably [45]. Therefore, AUC remains the more appropriate metric for comparing kernel sizes.

Figure 15 shows the absolute reconstruction residual for a −20 dB defective trace at an arrival time of 0.344 ms, using a kernel size of 123. For most of the signal, the residual remains low, indicating that the autoencoder reconstructs these regions of the trace well. A localised increase in residual is visible at the defect arrival time, showing that the CNN autoencoder reconstructed the defect region less accurately than the surrounding healthy regions. Large residual peaks are also visible at the beginning and end of the trace. These are likely caused by boundary effects introduced by zero padding at each convolutional layer [49]. This means that the convolutional filters have less realistic context near the signal boundaries than in the central region of the trace, causing reconstruction accuracy to be reduced near the boundaries and producing larger residuals [50]. This is a potential issue for defect detection, as the boundary residual on the right edge is larger than the residual associated with the defect response. This means that if residual magnitude was used as the anomaly detector, these boundary effects could increase the chance of false positives. A possible improvement to this would be using reflection padding instead of zero padding [50].

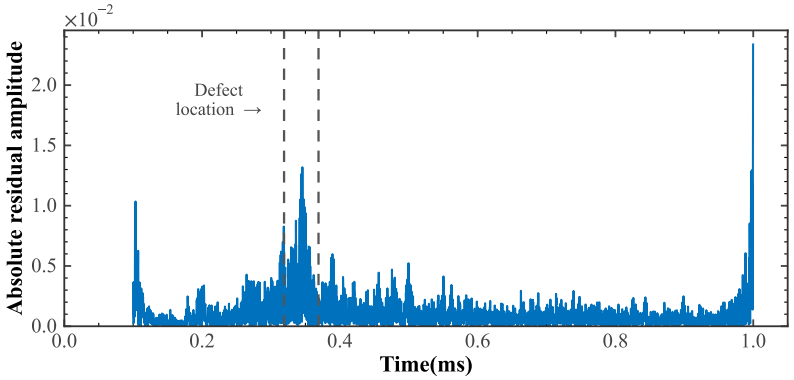


Figure 15: Absolute reconstruction residual for a −20 dB defective trace at an arrival time of 0.344 ms, reconstructed using the CNN autoencoder with a kernel size of 123. The dashed lines represent a 0.05 ms window centred on the defect. The larger peaks at the signal boundaries indicate edge effects from zero padding.

Reflection padding mirrors the signal values adjacent to the boundary during encoding rather than adding zeros, providing the model with a more realistic continuation of the waveform at the edges rather than the abrupt transition to zero introduced by zero padding [51].

4.3 LSTM Hyperparameter Studies (ES)

For the LSTM architecture described in Section 2.6, several hyperparameters were investigated before selecting an effective final model configuration. These included stride, training set size, history window length, and hidden size. An initial history window of length $k = 20$ or 0.004 ms was selected as a physically motivated starting point. Given that the emitted toneburst consisted of five cycles over 100 samples (0.02 ms), the chosen window length corresponded to approximately one cycle of the waveform. This provided local temporal context while remaining computationally efficient. An initial hidden size of 256 was selected and later refined through a dedicated parameter study. A batch size of 256 was also chosen to improve computational efficiency by allowing many windows to be processed in parallel during training. For all following studies, the model was trained using the parameters described in the reproducibility Table 8 in Appendix A.

As discussed in Tu et al. [21], MSE alone is not a sufficient metric for consistent defect detection in a single-step predictor. Although MSE was still used as the training objective for the hyperparameter studies performed in this section, models were compared primarily using AUC. To calculate this AUC, a benchmark set of 200 simulated defective traces with an amplitude of −30 dB and an arrival time of 0.384 ms was generated. The 200 benchmark healthy traces were selected from five equally spaced bins across the validation split. This stratified sampling aimed to select a representative range of traces spanning the full range of EOCs from the available data. For each subsequent study, five independent models were trained using different starting seeds. By using multiple starting seeds and calculating a mean AUC, each hyperparameter study could be initially assessed for stability and robustness to random parameter initialisation, and a consistent detector was more likely to be selected.

4.3.1 Stride study

For residual reconstruction, sliding windows are shifted with a stride of 1, so that every sample within a trace receives a prediction. For training however, using a stride of 1 for all windows would produce a very large number of highly overlapping windows, substantially increasing the computational cost with limited additional amplitude information. The purpose of this study was to reduce residual artefacts in healthy reconstructions, whilst ensuring an accurate prediction model that did not generate an excessive number of training windows. This study was performed prior to the other hyperparameter studies, as

subsequent studies aimed to compare AUC values between trained models and therefore required small and consistent residuals to ensure effective defect detection.

Initial reconstructions showed that a uniform global training stride did not provide sufficient residual consistency in the initial pulse arrival region. In Figure 16a, there is a visibly large peak within this arrival window and a near-zero residual elsewhere, indicating that the region most difficult to reconstruct was situated in the arrival window. This motivated the use of denser training-window sampling within this region. A training subset of 200 traces was used for this stride study, and a global training stride of 25 samples, corresponding to approximately one quarter of the 100-sample toneburst, was selected as a practical starting point. A focused arrival-window study was then performed.

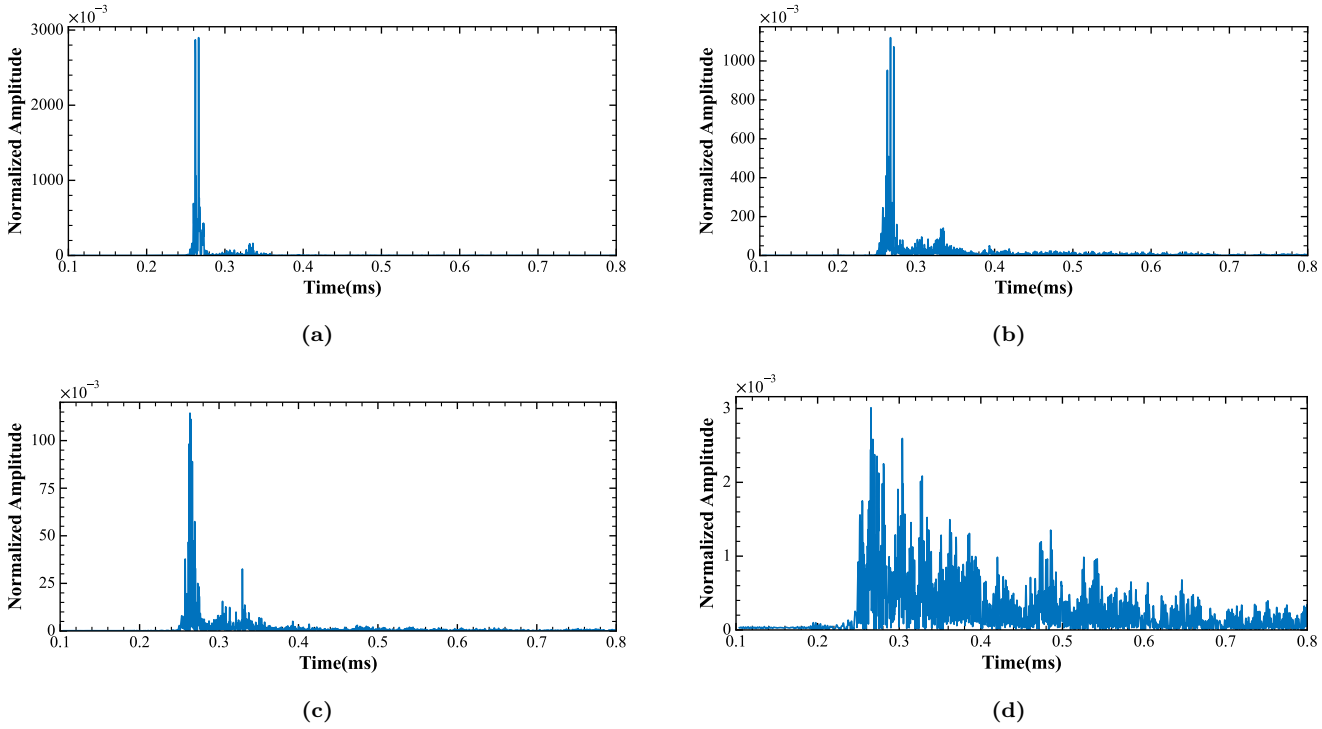


Figure 16: Healthy residual reconstructions for different focus-window stride strategies: (a) whole trace stride = 25 samples, (b) focus window stride = 10 samples, (c) 5 samples, and (d) 1 sample. Stride is the number of samples that are skipped before the next history window is taken. Absolute amplitude decreases as focus stride decreases, as seen by the y-scale reduction.

Focused strides of 10, 5 and 1 were examined within an arrival window of 500 samples, centred at the 1000th sample, corresponding to the 1500th sample in the uncropped trace. For each stride, a set of 20 residuals was reconstructed, with selected examples displayed in Figure 16. The most effective focused stride was 1, where the largest reduction in peak amplitude and a more consistent residual were observed. The average percentage reduction in peak amplitude between the 20 residuals of focused stride length 1 and the global stride of 25 was 99.75%, demonstrating a substantial reduction in healthy residual peak amplitude. Although this strategy increased training density locally, it did not change the reconstruction procedure used during evaluation or aim to reduce global validation MSE. Based on this improvement in healthy residual quality, the focused-stride training strategy was retained for all subsequent studies, in which model comparison was based on benchmark AUC.

4.3.2 Subset size study

Following the development of the focused-stride training strategy, a training subset size study was conducted to assess the data requirements of the LSTM model. In general, LSTM models require more

training data than NARX models because of their substantially larger number of trainable parameters. For example, the initial LSTM configuration used in this study contained 265,473 trainable parameters, compared with 1,721 reported for the NARX model of Wang et al. [3], reflecting the greater model capacity and correspondingly higher data requirements of the LSTM.

To ensure that each training subset captured a representative range of EOCs, the healthy training and validation data were divided into five sequential bins each, from which equal numbers of traces were sampled, as discussed previously. To calculate validation loss during training, a fixed validation subset of 250 traces was sampled across the five validation bins and used for all subset sizes, ensuring a consistent validation comparison. To select an appropriate training subset size, training time and AUC were compared across five training subset sizes, as shown in Figure 17a. The AUC was calculated from the ROC curve using the benchmark defective dataset and 200 healthy validation traces sampled across the five validation bins.

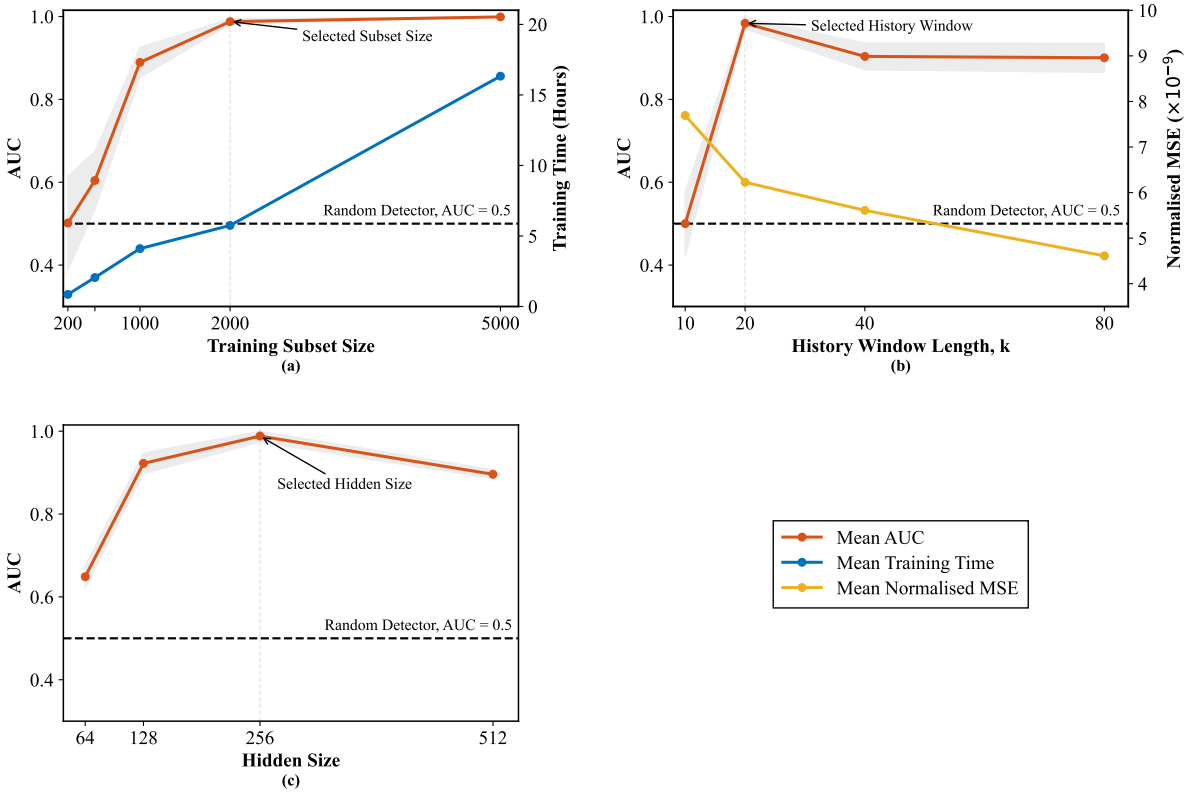


Figure 17: Banded plots of mean AUC with grey bands of ±1 standard deviation for (a) the subset size study, (b) the history window study, and (c) the hidden size study.

As shown in Figure 17a, training time increased nearly linearly with subset size, with 200 traces requiring only 50 minutes, or 3,085 seconds, to train, whereas 5,000 traces required approximately 16.5 hours. A clear trade-off was observed between mean training time and AUC, particularly between subset sizes of 2,000 and 5,000, where the mean AUC increased by only 0.007 while the training time increased by 11 hours. For this reason, a subset size of 5,000 was not taken forward into the remaining hyperparameter optimisation studies. To assess whether the observed performance was robust to random initialisation, the standard deviation across the five independently trained models with different random initialisation seeds is displayed as a grey band in Figure 17a, where a lower standard deviation suggests a more robust trained model. On this basis, the subset size of 2,000 was selected because it achieved highly converged, near-perfect defect detection with significantly lower computational cost than the 5,000-trace case.

4.3.3 History window study

The next study examined the effect of history window length on model performance. The history window length, k, controls how many previous samples are provided to the LSTM when predicting the next sample, thereby determining how much prior waveform information is available to the model. If k is too small, the model may not receive sufficient waveform information to capture the important temporal dependencies. Conversely, if k is too large, the model may be given redundant information, increasing computational cost without providing a corresponding improvement in defect-sensitive residual reconstruction. The aim was therefore to identify a history window length that provided the best balance between validation MSE, benchmark AUC and computational cost. This study allowed the relationship between the training objective of minimising next-step prediction error and defect-detection performance, measured through AUC, to be assessed directly.

Several history window lengths were tested while all other parameters were held constant, including the focused-stride training strategy, training subset size, optimiser settings and hidden size. For each value of k, five independently trained models with different random initialisation seeds were used to assess AUC convergence and robustness to random initialisation. For each seed and value of k, the training time, best validation MSE and benchmark AUC were obtained using the benchmark defective dataset. The mean validation MSE and mean AUC were then calculated and displayed in Figure 17b.

Reducing k below 20 resulted in a significant drop in AUC of nearly 50%, while the validation MSE remained comparatively stable. This suggests that, for shorter history lengths, the model could still minimise one-step prediction error using local continuity in the waveform, but was less effective when learning the waveform dependencies required to produce defect-sensitive residuals. This supports the interpretation that the LSTM can exploit meaningful temporal waveform dependencies beyond immediate local extrapolation. For values of k greater than 20, the AUC decreased by approximately 0.05 while training time increased substantially, indicating that increasing the available previous samples did not improve defect detection and instead increased model complexity unnecessarily. The results of this study suggest that approximately one cycle of the emitted toneburst provided sufficient temporal context, and the initially selected value of $k = 20$ was therefore retained for the remaining hyperparameter optimisation studies.

4.3.4 Hidden size study

The final hyperparameter study assessed the effect of varying the hidden size, H, of the LSTM. Although a higher hidden size may allow the model to capture more complex dependencies within the waveform, it increases computational cost and training time, as well as the risk of overfitting to the training data. Conversely, a lower hidden size restricts the model to a lower-dimensional recurrent feature representation, but can reduce training time significantly if the important dependencies do not require complex feature representation. This study aimed to identify a hidden size that allowed a sufficient representation of temporal waveform information while minimising computational cost.

Five random initialisation seeds were used for each hidden size of $H = 64$, 128, 256 and 512 to produce validation MSE, benchmark AUC and training time. From the five trained models, the mean AUC was calculated for each hidden size, and the standard deviation across seeds was compared in a banded plot, as shown in Figure 17c. The initial hidden size of $H = 256$ provided the highest AUC with effective convergence, suggesting that the chosen hidden size was sufficient to capture the relevant temporal structure of the waveform without requiring further model capacity.

Smaller hidden sizes, particularly $H = 64$, produced lower AUC values of up to a 35% reduction, suggesting that the reduced feature representation was insufficient to capture important waveform structure required for reliable defect detection. A hidden size of $H = 512$ increased model complexity but reduced the benchmark AUC to approximately 0.9, indicating that the additional model capacity provided no clear

detection advantage. A hidden size of $H = 256$ was therefore selected for the final trained model due to its highly consistent, near-perfect defect detection across all tested initialisations.

5 Synthetic Defect Detection Results (ES, OD, RJ)

To assess the detection performance of the three models, synthetic defective traces $s_{\text{syn}}(t)$ with known defect amplitudes β and arrival times t_d, and their healthy counterparts from s^{2013} were used. The formation of the synthetic defect dataset is described in Section 3.3. This allowed the selected models to be assessed across an extensive range of defect scenarios and enabled their detection sensitivity and consistency to be evaluated. The models were assessed using ROC curves and the corresponding AUC, as detailed in Section 2.7, where a threshold sweep over the residuals determined the defect-detection performance of each model. Before directly comparing the models, it is important to understand the two different threshold sweep strategies described in Section 2.7. An arrival-window strategy for OBS and a whole-trace sweep for the CNN and LSTM were used, so ROC and AUC comparison should be made cautiously.

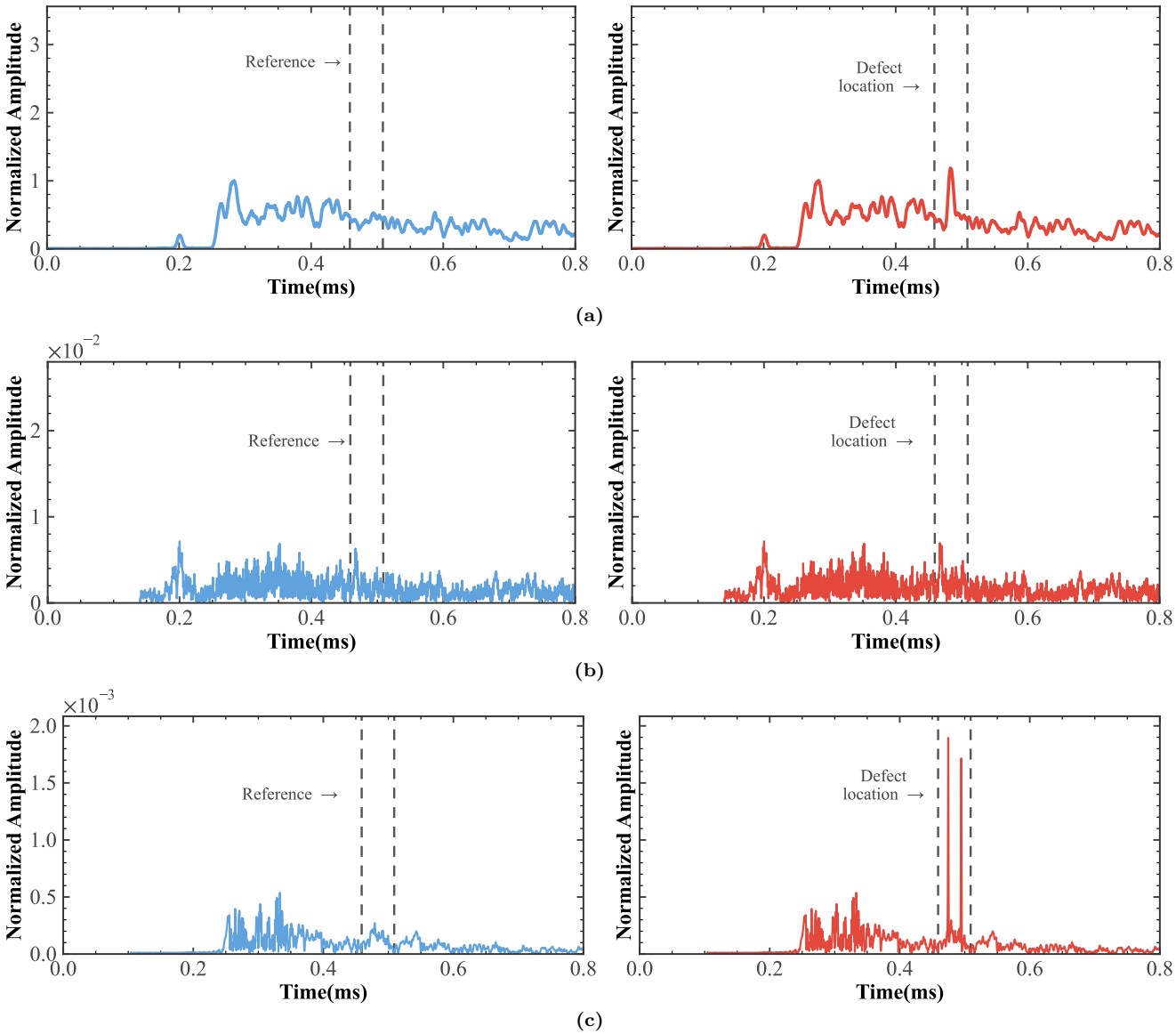


Figure 18: Comparison of example healthy and defective residuals for a defect with amplitude $\beta = -25$ dB and arrival time $t_d = 0.484$ ms for the (a) OBS, (b) CNN, and (c) LSTM models.

To visualise the reconstruction quality and subsequent residuals of the three models, a residual comparison was performed. Figure 9 displayed the healthy and superposed defect of arrival time $t_d = 0.484$ ms and amplitude $\beta = -25$ dB used in this comparison. Figure 18 shows the side by side comparison of this healthy and defective residual for the three models. Figure 18 shows limited visual separation between the healthy and defective residuals for OBS and CNN, indicating that a threshold-based residual sweep may struggle to detect defects for these models in this example case. In contrast, the LSTM shows a clear discrepancy between the healthy and defective residuals, indicating improved separation and making successful defect classification through threshold sweeping more likely. This figure only displays a single example residual pair, so it is used purely as a qualitative visual comparison. Quantitative performance was assessed through ROC curve generation using 200 healthy and 200 defective residuals for each arrival-time and amplitude combination. Figure 19 shows the comparison ROC curves for the five defect amplitudes at an example arrival time of 0.344 ms for the three models.

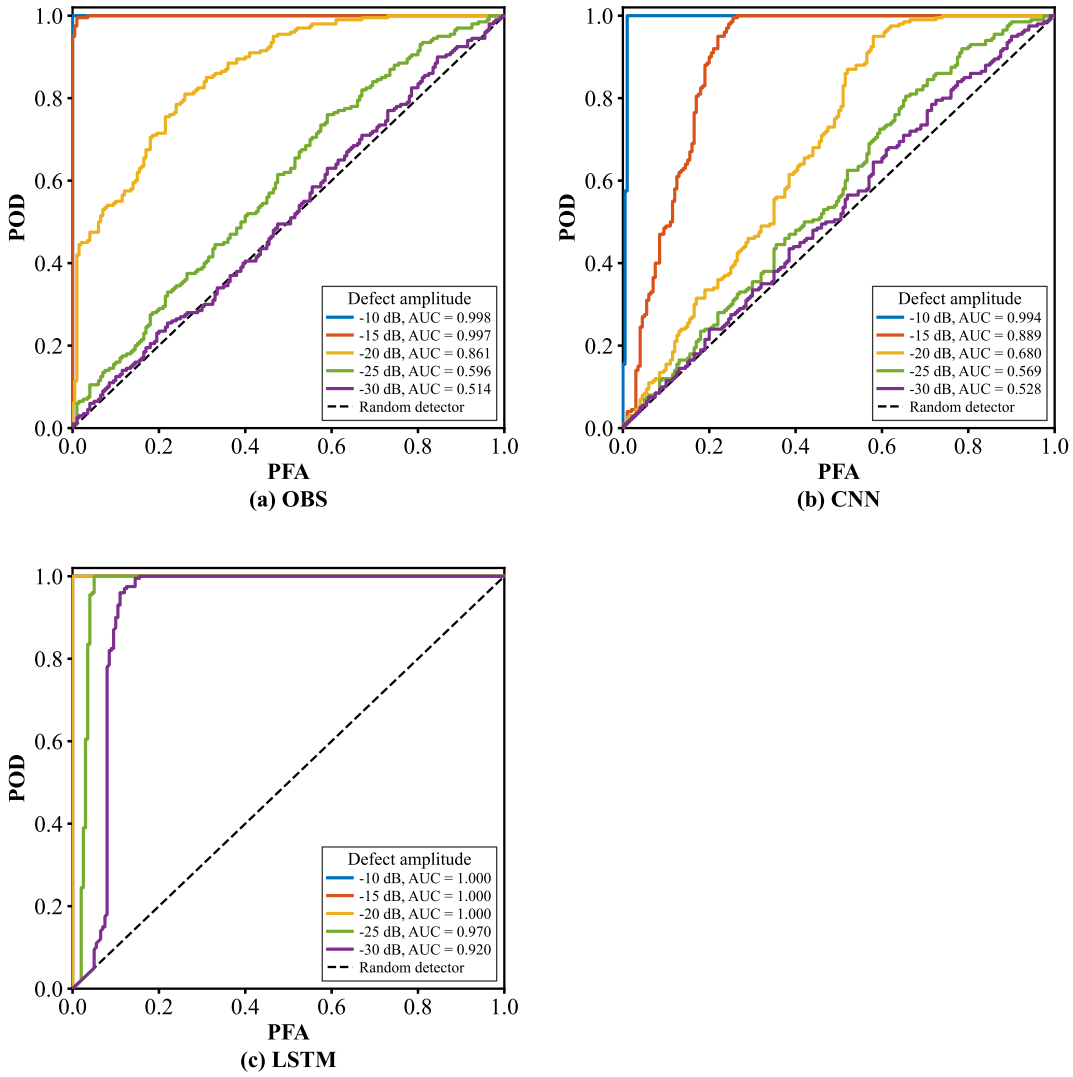


Figure 19: ROC curves for arrival time $t_d = 0.344ms$ for (a) OBS, (b) CNN, and (c) LSTM. Where POD is Probability of Detection, PFA is Probability of False Alarm, and ROC is Receiver Operating Characteristic.

Figure 19 shows that OBS achieved high separability for larger-amplitude defective signals, with AUC values close to 1.0, but performance decreased below $\beta = -20$ dB. For $\beta = -25$ and -30 dB, OBS approached random classification, yielding AUC values of 0.596 and 0.514, respectively. At this arrival time, the CNN produced progressively lower AUC values as defect amplitude decreased and performed less

successfully than OBS, approaching random classification for lower-amplitude defects. In contrast, the LSTM achieved perfect separability, with AUC = 1.0, for all defects of amplitude $\beta \geq -20$ dB. Although detection performance decreased slightly for $\beta = -25$ and -30 dB, the LSTM still retained AUC values above 0.9. For this arrival time, the LSTM therefore provided the strongest defect separability. The final synthetic-data evaluation extended the ROC/AUC analysis across all tested defect amplitudes and arrival times, with the calculated AUC values shown in Figure 20.

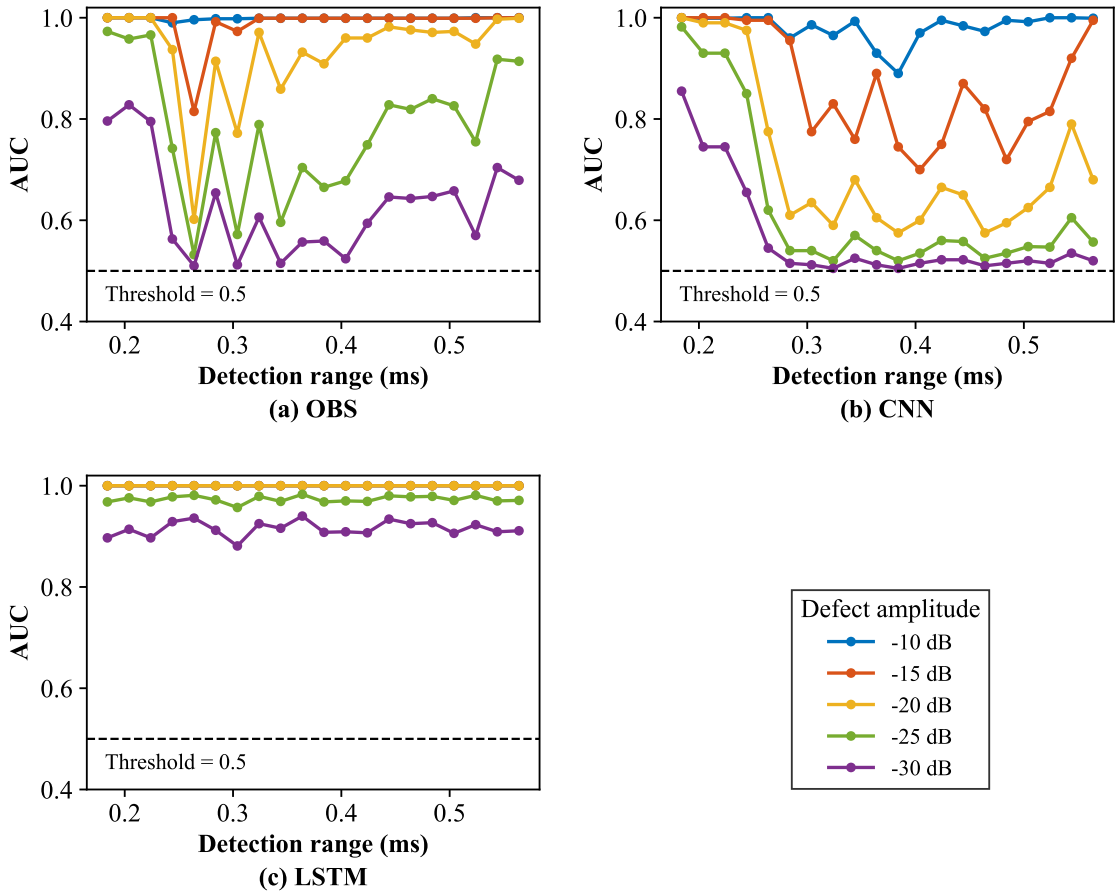


Figure 20: AUC comparison across all tested defect arrival times and amplitudes for (a) OBS, (b) CNN, and (c) LSTM models. The LSTM maintains high AUC for weak defects across the tested detection range, whereas OBS and CNN performance deteriorate substantially at lower defect amplitudes.

Figure 20 shows the full sweep of AUC values across arrival times and amplitudes for the three models. The OBS and CNN can be seen to struggle with lower defect amplitudes, $\beta \leq -25$ dB, whilst the LSTM provides consistent and sensitive detection performance. Table 5 gives a summary of the defect detection performance for the weakest and most difficult to detect defect amplitude, $\beta = -30$ dB, in which the greatest discrepancies between the performance of the three models was observed.

Table 5: Summary of synthetic defect detection performance evaluated using AUC for the weakest tested defect amplitude, $\beta = -30$ dB, across all arrival times. The LSTM is shown to provide the most consistent and sensitive detection across all metrics.

Model	Threshold strategy	Mean AUC at -30 dB	Worst-case AUC at -30 dB	AUC range at -30 dB	Main limitation observed
OBS	Arrival-window	0.6280	0.5100	0.3180	Strong arrival time dependence; detectability for weak defects degrades
CNN	Whole-trace	0.5653	0.5050	0.3500	No consistent improvement over OBS; weak defect residuals not reliably separable
LSTM	Whole-trace	0.9153	0.8810	0.0590	Occasional residual spikes, but strongest overall weak defect separability

6 Experimental Testing (BT, LH)

(BT)

The synthetic defect study provided a controlled test for the models, whilst the following experimental study aimed to generate real defect data and assess the models capabilities for defect localisation. The experimental phase of the project was designed to fulfil three main objectives:

1. To gain a more comprehensive understanding of GWSHM physics
2. To acquire defect-state data absent from the initial water tank dataset for validation training of the ML models in future works
3. To compare DAS and probabilistic data fusion damage localisation methods

Data acquired from the experimental setup needed strict comparability to the earlier setup for feasible implementation into the ML models. Consequently, the design was implemented under the same premise: a GWSHM system utilising piezoelectric transducers in a pitch-catch configuration to actuate and receive Lamb waves for the detection of structural defects, using a 5-cycle Hanning windowed toneburst as the excitation signal, and exciting the S_0 mode selectively.

6.1 Introduction (LH)

In order to supplement the previous synthetic defect analysis and computational model comparison, this study conducted a controlled experiment using an aluminium plate with a PZT sensor network. The purpose of this experiment was not to reproduce a specific in-service defect, but to generate repeatable defect-like disturbances under known laboratory conditions. The measured guided-wave data obtained from this experiment can be used to evaluate whether the selected signal-processing and localisation methods can respond to real physical disturbances, rather than only to artificially superimposed synthetic defect signals.

The aluminum plate was put between a pair of magnets to provide a localized mass-loading and a clamp-like disturbance. The influence of the location of the disturbance on the guided-wave response was investigated by moving this pair of magnets between pre-defined places in the central monitoring region. The same acquisition process and hardware configuration was followed throughout the experiment so that the signal variations at different known points could be compared.

Therefore, the main objective of this experimental phase is to establish a reliable and precise measurement

dataset, which will subsequently serve as the basis for comparing various signal processing methods. More specifically, this experiment aims to evaluate the effectiveness of each method in identifying the position-dependent variations of guided wave signals. It also aims to determine their respective sensitivities, consistency, and practicality under the same physical testing conditions. Essentially, this experimental work fills the gap between the physical sensing system and the subsequent results chapter, in which the accuracy of the methods and the overall applicability of the setup and selected materials are rigorously evaluated.

6.2 Methodology (BT, LH)

The methodology pipeline is shown in Figure 21, and detailed in the following section.



Figure 21: process of methodology

6.2.1 Setup (LH)

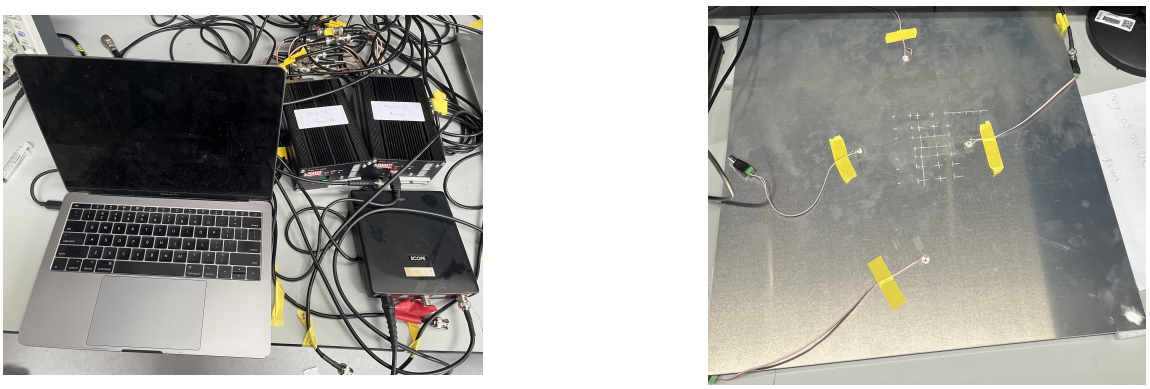
The experimental device consists of an aluminum plate and eight circular PZT sensors pasted on the surface. These PZTs constitute a Lamb wave sensing network, in which the selected PZT acts as the excitation end and the receiving end respectively to measure the guide wave response on different propagation paths. Signal generation and acquisition are controlled by TiePie HS3 Handyscope, multiplexer and personal computer. HS3 Handyscope is also used as a signal generator and oscilloscope, while the multiplexer is used to switch between different PZT excitation-receive combinations.

A pair of magnets are positioned on either side of the metal plate to create controlled local disruptions. The pair of magnets is moved between the preset settings of the 5 x 5 grid in the central monitoring area, with 25 test positions. The identical acquisition process and hardware configuration are used for each magnet point in the experiment. This can guarantee the comparability of the variations of the measured guiding wave signals under different known disturbance positions.

6.2.2 Materials and equipment (BT)

The piezoelectric transducers used in the experiment were Pz27 supplied from Ferroperm Piezoceramics A/S, with a 10 mm diameter and 1 mm thickness. The PZT wafers were bonded to the surface of two square isotropic 6082T6 aluminium plates with 500 mm length and 4 mm thickness using a superglue adhesive, assumed to cause complete mechanical coupling. Artificial non-permanent surface defects were introduced using N42 neodymium magnets, with 15 mm diameter and 5 mm thickness, placed on directly opposing positions of the aluminium plate, altering the local mass and causing incident Lamb waves slow in a manner comparable to the effect of a genuine structural defect. A TiePie Engineering HS3-Handyscope was used as a digital oscilloscope and signal generator, and multiplexor circuits built by the University of Bristol were used to physically switch between PZT wafer connections for automatic actuator-sensor cycling. The multiplexor and HS3 Handyscope are shown in Figure 22(a), and a plate showing the preliminary setup is shown in Figure 22(b).

The experimental setup utilised the exact same excitation signal as the water tank setup, explained in Section 3.1.3.



(a) **(b)**

Figure 22: (a)Experimental equipment used, including the PC, multiplexors and HS3 Handyscopesetup and (b)Setup used in preliminary experiment

6.2.3 Excitation frequency (BT)

Suitable excitation frequency selection was vital to elicit the desired system response. Tuning frequency refers to the frequency at which PZT wafers selectively excite desired modes [52]. Given that an ideally surface-bonded piezo transfers mechanical energy into the structure via shear forces at its edges, opposing points in a PZT wafer will thus generate waves that interfere; at the tuning frequency, a mode undergoes constructive interference for a significant increase in amplitude when the waves from opposing edges are completely in phase. This effect can also be reversed to completely suppress a given mode, which is achieved by selecting a frequency at which waves generated from the opposing edges are completely out of phase causing complete destructive interference. Given that the radial displacement of the top surface of the plate is mathematically governed by a Bessel function of the first kind [52], the roots of the function and the dispersion curves can be used to determine the optimal frequency at which A_0 is theoretically suppressed. Piezo resonance frequency occurs when the electrical input frequency matches the natural mechanical frequency of the piezo, causing electromechanical resonance where the physical displacement of piezo is at its maximum. To selectively excite the S_0 mode, the resonance frequency for the radial mode of the piezos can be calculated using a constant given by the manufacturers.

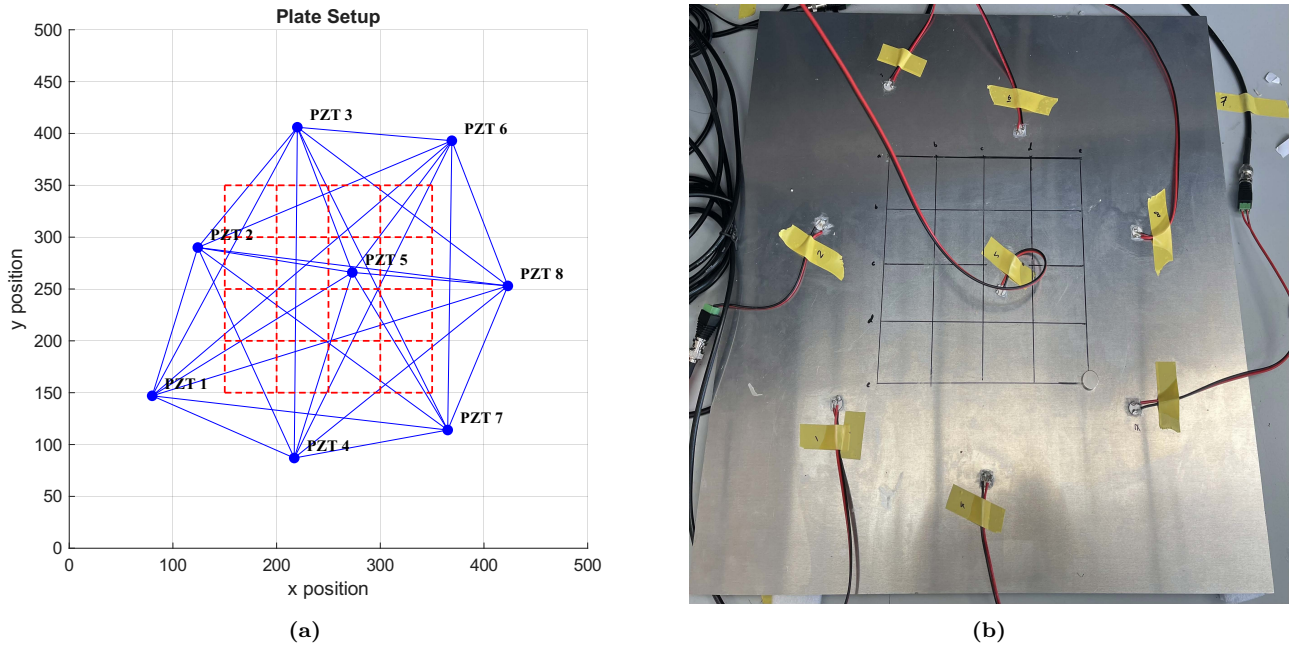
6.2.4 Preliminary experiment (BT)

Preliminary experiments were carried out with the purpose of establishing key characteristics required for the main defect data collection experiment. Two PZT pairs were bonded to the surface of the aluminium plate, as shown in Figure 22b, and the direct path between each pair (henceforth referred to as the sensing path) was evaluated. Firstly, at room temperature, a frequency sweep was conducted to determine the optimal excitation frequency at which the S_0 mode dominated the pristine response signal. Frequencies between 150 kHz and 350 kHz at 50 kHz intervals were considered, as well as the calculated tuning and resonance frequencies. The upper limit of 350 kHz was selected such that the frequency-thickness product did not exceed the threshold for higher order mode excitation. Next, to investigate the sensing path coverage, magnets were then positioned at varying perpendicular distances from the midpoint of the direct sensing path, starting from 10 mm and moving outward in 10 mm intervals until the artificial defect could not be clearly identified. The setup used in the preliminary experiment is shown in Figure 22(b).

6.2.5 Data collection experiment (BT)

The main experiment aimed to collect data on defect-state response signals. Adapting the pitch-catch sensor layout from the water tank to the physical dimensions of the experimental plate presented a geometric challenge. While a direct spatial scaling of the original sensor array was initially proposed,

this approach caused insufficient mode separation and was discarded. Thus an alternate sparse array was designed by evaluating each actuator-sensor pair and applying two constraints: First, the direct propagation distance x between each pair must be greater than a minimum threshold D to achieve sufficient mode separation, expressed as the inequality $x > D$. Second, the difference between the shortest reflected path R and x must be greater than a minimum path length difference value P, such that reflected signals do not interfere with the arrival of the first mode, expressed using the inequality $R - x > P$. Using the S_0 and A_0 group velocities, and the time period for one complete transmission of the excitation signal, D and P can be determined. Thus applying the formulated constraint expressions into the constrained nonlinear multivariable minimisation function fmincon in MATLAB gives the proposed arrangement in Figure 23(a). The physical setup is also shown in Figure 23(b).



(a) **(b)**

Figure 23: Experimental setup use in data collection experiment, (a)plotted in MATLAB, showing sensing paths and grid of defect locations and (b)the physical setup

This approach was chosen over conventional setups due to several considerations: Conventional setups often feature PZT wafers in a symmetric rectangular shape, often near or on the edges of the plate. Redundant sensing paths caused by collinear PZT wafers lead to suboptimal data utilisation, and boundary reflections causing complex signal interference. While baseline subtraction could be used to mitigate signal interference, the need for compatibility with the ML models necessitated a setup with wave packets that arrived before interference. The proposed setup ensures that all sensing paths are unique and wave packets arrive before complex boundary reflection.

6.2.6 Data Processing (BT)

Data processing techniques were applied to the data collected before subsequent analysis. To account for shifts in the time domain caused by minute changes in ambient temperature, a Baseline Signal Stretch (BSS) technique [36] was implemented such that baseline subtraction yielded comparable residual graphs where the effects of EOV were minimised. The BSS technique implemented stretches the time-axis of a baseline signal to match the response signal, theoretically eliminating any phase or frequency shift, and is detailed in Section 4.1.2. Additionally, recorded signals experienced high interference from boundary reflections after the arrival of the first S_0 packet, indicating that isolation of said packet would be ideal for analysis. Thus the signal was truncated using a 0.022 ms window centred on first S_0 peak, with the window period defined using the original excitation period given an additional 10% margin. A bandpass

filter centred on the excitation frequency with bandwidth 100 kHz was also applied to filter out unwanted high and low frequency noise, and the DC offset was eliminated in the trace such that the traces were centred vertically on the y-axis.

6.2.7 Defect detection and localisation methodology (BT)

To characterise the deviation of a system response signal compared to a baseline signal, multiple Deviation Indices (DI) were considered to numerically quantify the signal. Firstly, the Root Mean Square Deviation (RMSD) is a DI that accounts for both amplitude variation and time shift [53], making it an ideal damage indicator for EOV sensitive Lamb waves. Secondly, the Amplitude Variation (AV) compares the difference in amplitudes of a selected peak, detecting the energy loss of a packet caused by the defect scattering incident Lamb waves. Finally, the Cosine Distance (CD) is a commonly used method to gauge the similarity of two vectors. Given that it utilises the dot product in its application, it is hypothesized to be adept at measuring phase and frequency shifts, while weak at detecting changes in amplitude. RMSD DI was predicted to be the most robust due to its flexibility, while the AV DI was predicted to be the least accurate due to its application to the first S_0 rather than a later arrival - a limitation imposed by boundary reflections. Applying the AV DI to the first packet was not ideal as the wave energy loss would not be detected. Additionally, the sensitivity of CD DI to frequency and phase shifts suggested that it would perform poorly due to the implementation of BSS, and that the magnets were not expected to change signal frequency or phase.

The RMSD DI was applied to both the full signal and the first S_0 packet, and is expressed in Equation 16,

$$RMSD = \sqrt{\frac{\sum_{i=1}^{N}(S_d(i) - S_b(i))^2}{\sum_{i=1}^{N} S_b(i)^2}} \tag{16}$$

where the index i is the ith data point in the signal, and S_d and S_b are the system response signal and baseline signal respectively. The AV DI was used to compare the amplitude differences of the first S_0 arrival packet, and is expressed in Equation 17,

$$AV = \left| 1 - \frac{amplitude(1^{st}peak(S_d))}{amplitude(1^{st}peak(S_b))} \right| \tag{17}$$

Finally, the CD DI was used to quantify the phase and frequency shift of the signals, as expressed in 18

$$CD = 1 - \left| \frac{S_b \cdot S_d'}{\sqrt{(S_b \cdot S_b')(S_d \cdot S_d')}} \right| \tag{18}$$

where S_b' and S_d' are the transposed signals of S_b and S_d respectively.

A data fusion method [53] was used to normalise the DIs for comparison. The path with the highest DI was used to normalise the DIs of all other sensing paths, such that the resulting DIs ranged from 0 to 1, least damaged to most damaged. The Total Anomaly (TA) was then calculated by taking the mean of every available normalised DI for every path.

The probabilistic damage localisation approach proposed by Wang et al. [54] and refined in [55] was utilised for imaging purposes. The area enclosed within the sensing paths, named the monitoring area, was discretised into uniform grids with 1 mm length. The probability of a defect at any given point within the monitoring area, $P(x, y)$, was then estimated using the expression as given by Equation 19:

$$P(x,y) = \sum_{j=1}^{J} P_j(x,y) = \sum_{j=1}^{J} TA_j f_j(z) \tag{19}$$

Where $P_j(x,y)$ is the probability of damage at (x,y) obtained from the deviation index from the jth path, TA_j is the TA of the jth path, and $f_j(z)$ is the normal distribution function for the jth path. The effect of the TA is modelled such that it is maximised when defects lie on the sensing path and varies with distance to the sensing path according to the normal distribution, as given by Equation 20:

$$f(z) = \frac{1}{(\sigma\sqrt{2\pi})} e^{-\frac{(z-\mu)^2}{2\sigma^2}} for -\infty < z < +\infty \tag{20}$$

Where μ is the mean of the normal distribution model, σ is the standard deviation, and z is the normal variable. μ was set to one and σ was set to the sensing path coverage previously determined in the preliminary experiments. The variable z was used to designate the weighting of a grid square according to some measure of distance from the sensing path, with closer grids being assigned a heavier weighting. Confining z within an elliptical zone using Equation 21, damage perception near the centre of the sensing path was improved, consequently improving damage perception of the monitoring area [55].

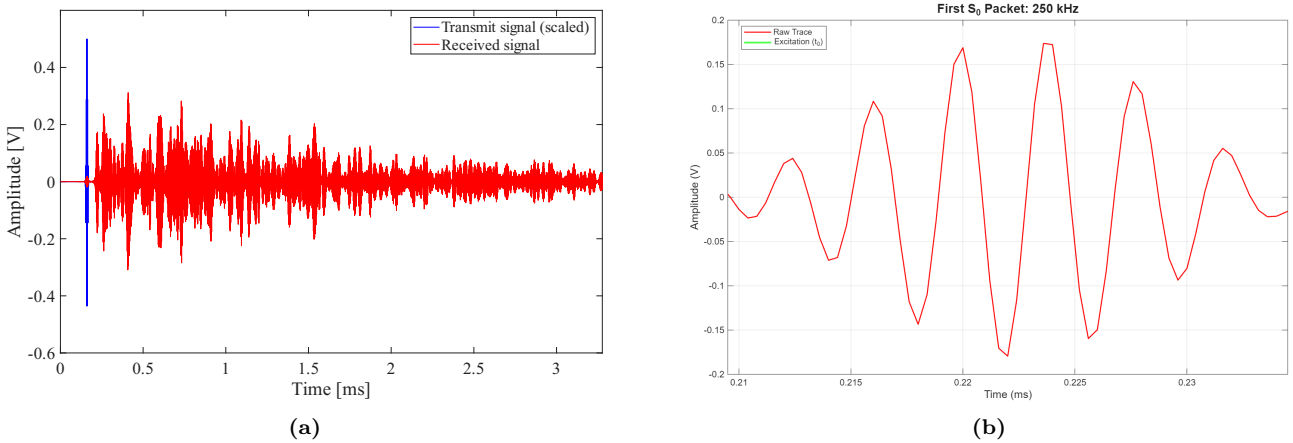
$$z = \frac{D_a + D_s}{D_k} \tag{21}$$

where D_a is the distance of the grid from the actuator, D_s is the distance of the grid from the sensor, and D_k is the distance between the actuator-sensor. Thus z represents a dimensionless ratio of distances, minimised when $D_a + D_s = D_k = 1$, at which point the grid point lies directly on the sensing path, maximising $f(z)$.

7 Experimental Results (BT, LH, OD)

7.1 Preliminary results (BT)

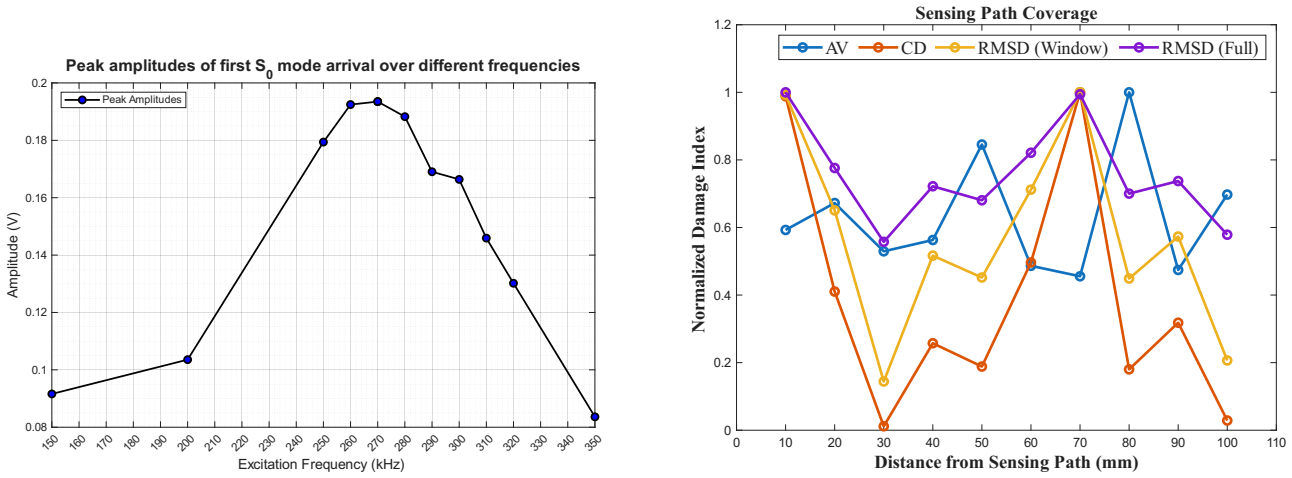
As previously stated, significant boundary reflections in the received signal necessitated isolation of the first S_0 packet, as demonstrated in Figure 24.



(a) **(b)**

Figure 24: (a)Trace of entire signal at 250 kHz and (b)the isolated S_0 packet, both acquired during the frequency sweep

The optimal A_0 suppression frequency and resonant radial frequency were calculated to be 290 kHz and 201 kHz respectively. Conducting the frequency sweep and plotting the amplitudes of each frequency in Figure 25(a), it can be seen that 270 kHz induced the highest amplitude for the first S_0 mode arrival at 0.19 V, and was thus selected as the excitation frequency, However, the peak amplitude of the A_0 packet could not be evaluated comprehensively due to overlap with boundary reflections. It should also be noted that the PZT wafers do not seem to be significantly excited at the radial frequency 200 kHz, with a peak amplitude of 0.1 V. Sensing path coverage testing was conducted by considering a PZT wafer pair 20 mm apart in the preliminary experiment. The DIs were then applied to investigate the maximum distance at which defects appear visible. The normalised DIs are plotted in Figure 25(b).



(a) (b)

Figure 25: Results of the tests conducted in the preliminary experiment, plotting (a)First S_0 peak amplitudes of different frequencies using the frequency sweep results, and (b)Damage detected against distance from sensing path using sensing path coverage testing results

Note that a higher number of frequencies were tested neighbouring that of the peak amplitude, as seen in Figure 25(a), located very near to the calculated tuning frequency.

Considering the trends present in Figure 25(b), maximum deviation occurs at 10 mm. The DIs then decrease until 30 mm, where all indices experience a local minimum, then exhibit a general upward trend, with all DIs except the AV DI spiking sharply at 70 mm, registering maximum deviation, before falling again. This matches perfectly with the location of the second PZT wafer pair(see Figure 22b(b)), indicating that there is a high sensitivity to the surface bonded PZT wafers and low sensitivity to the magnets due to the deviation from the PZT wafer having the same value as that caused by the magnet. The presence of the deviation caused by the second PZT wafer pair despite baseline subtraction and BSS processing also indicates flaws within the methodology, and the trend-resistant and chaotic nature of the AV DI also implies its unsuitability in this scenario. The trends suggest that the system was incredibly sensitive to defects only when positioned directly on the sensing path, while simultaneously highly sensitive to PZT wafers even when they are far from the sensing path, evidenced by the two points of maximum deviation. Thus, according to these results, a sensing path coverage distance of 30 mm was selected, as this is the distance at which the deviation caused by the magnet is at its minimum.

7.2 Defect localisation results (BT, OD)

7.2.1 Delay-and-sum (OD)

To process the experimentally acquired data, the concepts discussed in Section 2.2 were extended into the delay-and-sum (DAS) method to produce localisation heat maps [56, 57], as shown in Figure 26.

Whilst the resulting plots show repeatable residual structure, they do not demonstrate reliable localisation of the known defects, as shown by the localisation error for all 6 magnet positions in Table 6. The implications of this error on the experimental setup are discussed in the subsequent sections.

Table 6: Defect localisation error from DAS plots for selected magnet positions. All numerical values shown are in mm.

Position	Expected X	Expected Y	Peak X	Peak Y	Total error
AA	150.0	350.0	472.0	54.0	**437.4**
BC	250.0	300.0	36.0	0.0	**368.5**
BE	350.0	300.0	472.0	54.0	**274.6**
CB	200.0	250.0	164.0	240.0	**37.4**
EB	200.0	150.0	472.0	54.0	**288.4**
EE	350.0	150.0	470.0	50.0	**156.2**

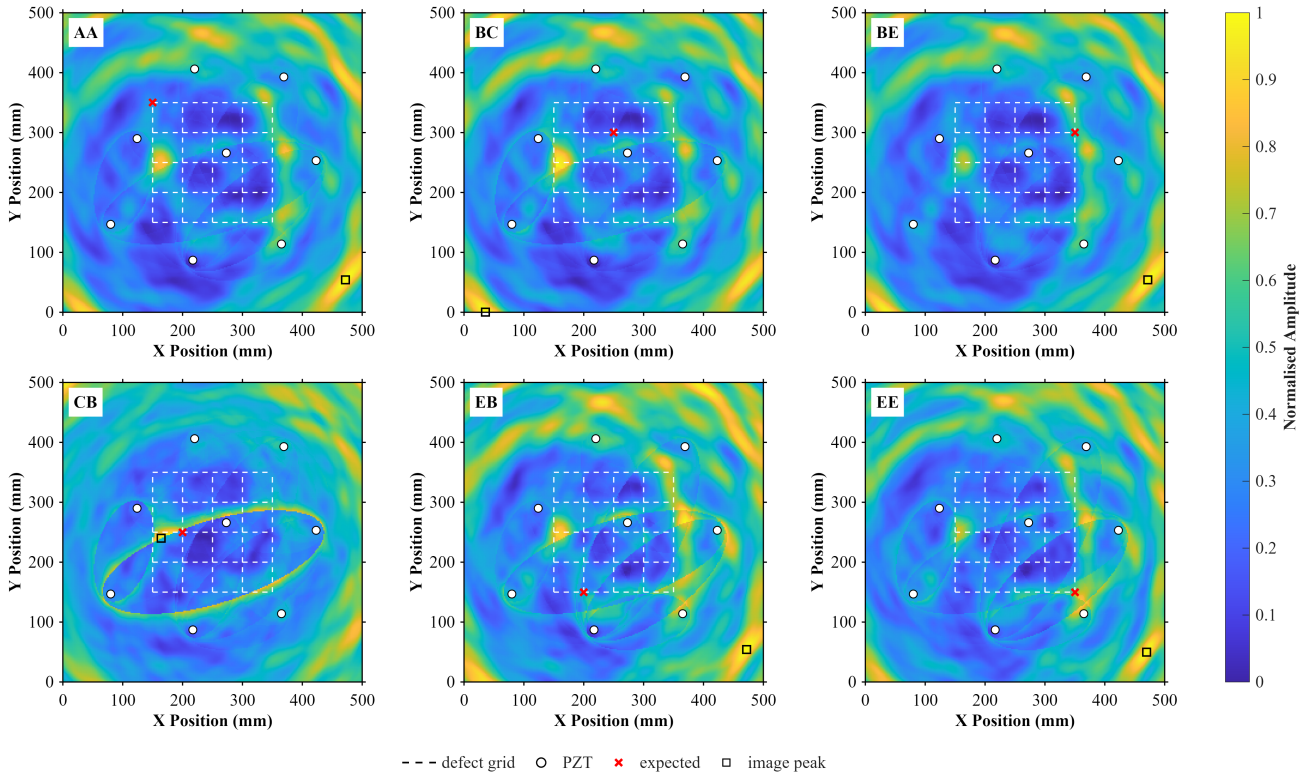


Figure 26: Delay-and-sum amplitude heat maps for six different defect positions: (a) AA, (b) BC, (c) BE, (d) CB, (e) EB and (d) EE. The intended defect grid is overlaid with the actual position of the magnet marked by a red cross.

7.2.2 Probabilistic data fusion (BT)

Similarly, using the probabilistic data fusion approach outlined in 6.2 did not yield reliable localisation results, as seen in Figure 27.

As observed in the heatmaps, it cannot be said that the magnets are accurately located. There appears to be no affect on the local deviation index - areas where the defects are positioned do not show a noticeable change in colour. Additionally, the entire monitoring area detects a high level of deviation and PZT wafer 5 appears to maintain the highest detected deviation across different defect positions, a highly unexpected result when considering that baseline subtraction should eliminate features that do not vary between measurements. Furthermore, while Figure 27(c) shows variation as compared to the other

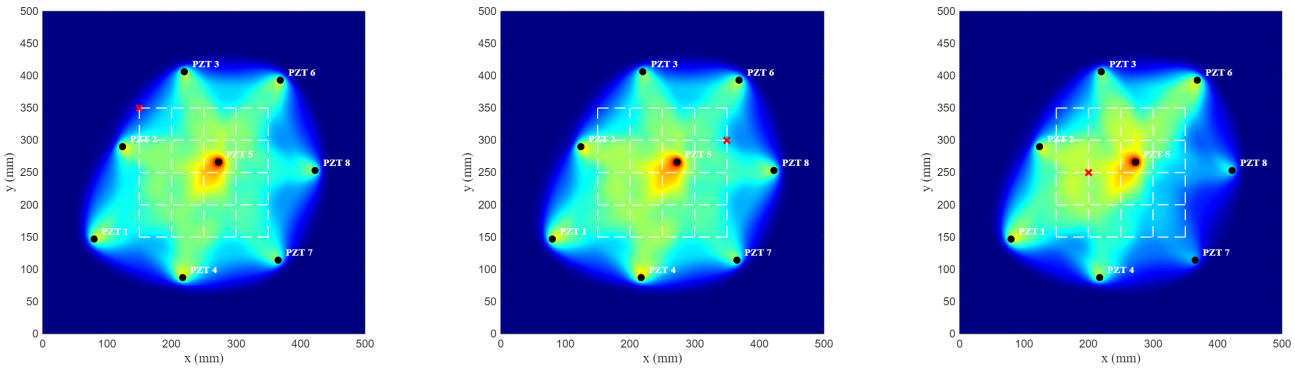


Figure 27: Probability based amplitude heatmap for different defect positions: (a) aa, (b) be, (c) cb, where defect positions are marked by a red cross on the grid

heatmaps, it is not consistent with predicted results and appears to be a random abnormality, attributed to the effects of EOV. Given that both localisation methods do not accurately locate the defect, it can be assumed that the fault lies within the experiment setup. It is theorised that this effect is caused by a multitude of factors: the magnets, the nature of BSS, DI superposition and the aluminium plate; limitations are further discussed in Section 8.

7.3 Accuracy study (LH)

Experimental localisation results show that the current experimental setup cannot reliably identify the real magnet position. The accuracy of this section is mainly by comparing the high-response regions predicted in the localisation maps with the known magnet posi on the 5 × 5 test grid. Tions to evaluate. Although heat maps produce measurable and repetitive deviation patterns, the peak responses do not coincide steadily with the actual magnet locations. According to Table 6, the best positioning effect is the CB position, with a localisation error of 37.4 mm; the largest error is the AA position, with an error of 437.4 mm. This shows that the strongest reconstructed responses are not mainly determined by magnet perturbation itself, but are obviously affected by boundary reflections, s The influence of ensing-path geometry, data-fusion effects and local sensitivity near surface-bonded PZT wafers. Therefore, the experiment can prove that the magnet pair does disturb the measured guided-wave signals, but under the current setup, the localisation accuracy is still limited.

8 Discussion (OD, ES, RJ, BT)

This section compares the LSTM and CNN against the OBS benchmark using the results presented in Section 5, alongside the limitations and broader further work for each part of the project. As hypothesised, the LSTM provided more sensitive and consistent defect detection, which was most apparent for the weakest defect amplitude, where the LSTM achieved a mean AUC of 0.9153 at -30 dB compared with 0.6280 for the OBS and 0.5653 for the CNN. This suggests that the LSTM reconstructed healthy waveform structure more effectively while leaving defect-induced components visible in the residual. Through its gated structure previously described, the LSTM was able to learn important features from previous waveform samples and predict its next sample accordingly. In comparison, OBS was limited by its reliance on a single best-matched baseline with global amplitude scaling and time dilation, which may not fully compensate for local waveform distortions caused by EOC variation. The CNN autoencoder was also less effective because its pooling and bottleneck structure may suppress small defect-related features during compression, while zero-padding boundary effects increased residual artefacts unrelated to damage. This comparison should be interpreted with caution because the OBS was evaluated using an arrival-window

threshold sweep, whereas the CNN and LSTM were evaluated using a whole-trace threshold sweep, as detailed in section 2.7.

The standardisation method described in Section 4.3 explains how amplitude information is preserved across traces during training. This may help the LSTM to learn some amplitude variation between signals and produce predictions potentially more robust to EOC-induced amplitude variation. Also, through its learned recurrent memory the LSTM is able to more flexibly represent feature information from past history samples than the fixed explicit lag structure previously used in the single-step NARX predictor used in Wang et al. [3]. This standardisation method and more adaptive memory may explain how the LSTM was able to achieve such consistent and sensitive defect detection, though comparisons to the NARX are made cautiously due to the fact the models were trained and assessed on different sensor pairs.

For the CNN autoencoder, several architecture choices may have limited performance. Repeated pooling and bottleneck compression could have suppressed small defect-related features before reconstruction, reducing residual comparison for weaker defects and weakening anomaly detection. As discussed in Section 4.2.1, zero-padding produced large residuals at the boundaries, so using alternative methods such as reflection padding may give the model more of a realistic continuation of the signal near the boundary. Further work could also investigate alternative convolutional structures. Section 4.2.1 showed that no kernel size performed optimally across all defect amplitudes and detection ranges. A multi-branched CNN architecture with parallel convolutional layers of differing kernel sizes could capture both localised and broader waveform structures simultaneously. Noisy results produced in the kernel study could have been attributed to random weight initialisation and early stopping. Repeats for each kernel size to find the mean and standard deviation could mitigate the influence of anomalous runs. Also, Figure 13b shows the AUC values had not shown clear signs of plateauing, suggesting that larger kernel sizes could be tested for improved performance. These methods were not performed in this study due to increasing computational demand in each.

Table 4 summarises some of the recent developments made in overcoming the inherent limitations of the traditional OBS model, which are described in section 4.1.4. Most notably, adaptive compensation for non-uniform temperature has been explored using local phase matching [37]. This bypasses one of the most significant limitations, the use of global temperature compensation, which is promising when considered alongside parallel work on up-scaling temperature compensation techniques [7]. These developments suggest OBS could remain significant beyond providing a benchmark method, especially given its simplicity and use of only healthy data.

The main limitation of the defect detection study on the water tank is that model performance was evaluated primarily using synthetic defects. This allowed controlled comparison across defect amplitudes and detection ranges, but the defect model was a simplified delayed and scaled toneburst superposed onto pristine data. Real damage may introduce additional scattering, mode conversion, shadowing and geometry-dependent effects that were not represented. The analysis was also limited to a single sensor pair, so the results demonstrate detection sensitivity rather than full defect localisation performance across a sensor network. Further, the internal condition of the structure was static due to the tank being empty throughout the monitoring period. To carry out robust testing that would support confidence in using an LSTM for damage detection in industrial applications, such as for pipelines and gas storage tanks, there would need to be consideration for variations in internal pressure and flow rate. This presents an opportunity for further study on the feasibility of the LSTM architecture for damage detection.

For the experimental study, it was hypothesized that the magnets were not an effective artificial defect, causing little to an unsubstantial effect on the response signal, supported by the analysis of the preliminary results and the failure of the damage localisation methods to detect defects. It is further suggested that the thickness of the plate caused a large enough reduction in magnetic clamping force that the chosen

S_0 mode interacted poorly, especially given its in-plane motion. Contrarily, the mechanical coupling of the PZT wafers were much stronger, resulting in a greater interference in the response signals. As such, when the DIs were normalised, any deviation caused by the magnets were scaled to an insignificant value, justifying the trend-resistance of AV DI - the poor mechancial coupling caused little scattering and negligible S_0 mode conversion into A_0, resulting in little amplitude loss. This also explains why the sensing paths are highlighted in the heatmaps - with no significant defect acting on the plate, the phase and frequency differences that BSS failed to eliminate dominate the DI normalisation process such that all sensing paths are considered to be defective. This failure of the BSS method is due to the assumption that the changes in EOCs are global, failing to account for local EOVs. Additionally, the consistency of PZT wafer 5 as the point of highest deviation is due to the data fusion method. As the DI of each sensing path is summed, the point with the most sensing path intersection naturally becomes the point of highest deviation. The poor defect selection also led to poor preliminary testing results, leading to a poor sensing path coverage value. The suitability of fmincon for PZT configuration is also be debatable. The algorithm can become trapped within local minima, leading to poorly optimised solutions. As such, other optimisations such as Genetic Algorithms might be more suitable. Finally, the aluminium plate can be blamed for many more of the experimental failings. The low attenuation of the plate caused by the low internal damping of the material exacerbates the weakness of Lamb waves to EOVs and also caused significant boundary reflections over different frequencies, limiting the number of packets that could be evaluated.

9 Conclusion (OD, ES)

This project explored the feasibility of implementing deep learning models for defect detection in the context of GWSHM. An LSTM predictor and a CNN encoder–decoder were selected as the trace reconstruction architectures to retain the practical advantage of the OBS approach, whereby training can be performed with a purely healthy dataset. It was hypothesised that data-driven trace prediction would produce more informative residuals and consequently, provide more sensitive and consistent defect detection than the whole trace scaling performed by OBS. Section 5 provides strong evidence that the LSTM is consistent with this hypothesis as the evaluated metrics demonstrate overall improvement of defect detection, especially for lower amplitude synthetic defects. Quantitatively, the LSTM showed a 0.2873 or 45.7% improvement over the OBS in mean AUC for the smallest assessed defect amplitude and a worst case improvement of 0.37 (72.7%). In contrast, the CNN showed limited improvement in detection sensitivity over the OBS, suggesting the architecture of the LSTM is more suited to next sample prediction.

Although this study has limitations, particularly the lack of integration of the experimental testing with the ML models, the preliminary qualitative comparison to the NARX predictor, described in Wang et al. [3], suggests that the LSTM provides improved consistency in defect detection and was more sensitive to lower amplitude defects, even when using an alternative threshold sweep strategy. Whilst these results are not directly quantitatively comparable, there is justification for further research into experimental testing and further comparison of the LSTM and the NARX models, as well as the potential of advancement into multi-step prediction as described by Tu et al [21]. Overall, the LSTM approach to trace prediction is a promising route for residual-based GWSHM, particularly when defective data is limited for training, and real defect locations are unknown.

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A Model Training and Hyperparameter Settings

This appendix summarises the main training settings used for the CNN autoencoder and LSTM predictor. These settings are provided for reproducibility, while the main methodological justification is retained in Sections 4.2 and 4.3.

Table 7: Summary of CNN autoencoder training and hyperparameter settings.

Setting	**Value**	**Purpose**
Input channels	Real component only	Reduced input complexity while retaining dominant defect-related waveform information.
Input length	4500 samples (0.9 ms)	Cropped signal region used for reconstruction.
Training traces	1500	Healthy traces used to train the autoencoder.
Validation traces	500	Healthy traces used for validation and early stopping.
Test traces	200 healthy and 200 synthetic defective traces	Used for residual generation and detection evaluation.
Optimiser	Adam	Adaptive optimiser used for stable neural-network training over Stochastic Gradient Descent (SGD) [58].
Initial learning rate	0.001	Starting learning rate for model optimisation.
Scheduler	ReduceLROnPlateau	Reduced learning rate when validation loss plateaued.
Scheduler settings	Factor = 0.5, patience = 5	Controlled learning-rate reduction during training.
Early stopping	Patience = 50 epochs	Stopped training when validation loss no longer improved.
Kernel sizes tested	5–159	Used to assess the effect of convolutional receptive field size.
Selected kernel size	123	Best-performing kernel size from the CNN study.

Table 8: Summary of LSTM predictor training and hyperparameter settings.

Setting	**Value**	**Purpose**
Input channels	Real component only	Single amplitude feature used for next-sample prediction.
Input length	4500 samples (0.9 ms)	Cropped trace region used for prediction and residual reconstruction.
History window, k	20 samples (0.004 ms)	Selected temporal context for each next-sample prediction.
Prediction horizon, h	1 sample	Single-step prediction formulation.
Hidden size, H	256	Selected recurrent feature representation size.
Number of LSTM layers	1	Single recurrent layer used for waveform prediction.
Batch size	256	Mini-batch size used during training.
Training subset size	2000 traces	Selected as the best performance/runtime compromise.
Validation subset size	250 traces	Fixed validation subset used during training.
Training stride	25 samples globally; 1 sample in focused arrival window	Reduced training cost while improving residual quality around the arrival region.
Optimiser	Adam	More adaptive optimiser used for stable training over Stochastic Gradient Descent (SGD) [58].
Initial learning rate	0.001	Starting learning rate for model optimisation.
Scheduler	ReduceLROnPlateau	Reduced learning rate when validation loss plateaued.
Scheduler settings	Factor = 0.25, patience = 12, minimum LR = 1×10^{-9}	Controlled learning-rate reduction during training.
Early stopping	Patience = 100 epochs	Stopped training after prolonged lack of validation improvement.
Gradient clipping	Applied	Used to improve recurrent training stability [30].
Model selection metric	Mean benchmark AUC over five seeds	Used because validation MSE alone did not reliably indicate defect separability.
Benchmark defect case	-30 dB at $t_d = 0.384$ ms	Used for hyperparameter comparison.